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Acrylonitrile Electrohydrodimerization (EHD) — 1D Planar Electrode Model

Bui Lab | NYU Tandon School of Engineering | April 2026 | Guide v7

Scope: 1D Nernst diffusion layer (δ), Nernst–Planck transport with migration, Tafel kinetics for ADPN/PN/HER/TCH, phosphate buffer chemistry (OH^--pathway), regime-aware multiphase effective diffusivity on a Cd cathode. **Operating temperature: T = 298.15 K (25 °C).**

v7 additions over v6 (see §22 changelog summary at the end of this document): TCH (1,3,6-tricyanohexane) added as the 9th species and 4th cathodic reaction (§2, §5); AN reaction orders `n_ADN`, `n_PN`, `n_TCH` are promoted from hardcoded constants to fit parameters via the `KIN_OVERRIDE` mechanism (§5); HER kinetics frozen at v6.x converged values (§20); fit dimension extended to N_THETA = 9 (§20). Carries forward all v6 features — cell-voltage decomposition (§17), hydrodynamic δ mapping (§18), the Bloomquist et al. (CEJ 2026) 162-row experimental dataset (§19), the transport-frozen / kinetics-only fitting strategy (§20), and the Core/Extended/Holdout row-selection convention.

Bubble-induced void corrections are still deferred to v8 — v7 model is expected to track Bloomquist's 0.5 mm and 1.0 mm gap data well but systematically underpredict the 0.25 mm gap (Holdout) FE_ADN and V_cell because the H_2 bubble convection enhancement of mass transfer is absent.

Convention

All concentrations c_i in this guide are **aqueous-phase** values [mol m^{-3} of aqueous solution]. Under the local-equilibrium assumption (Da » 1), organic and aqueous phases equilibrate instantaneously at every position: c_i,org(x) = m_i × c_i,aq(x). The organic phase acts as a parallel transport pathway captured entirely through the effective diffusivity D_i,mix. **No explicit phase transfer term R_PT appears** in the governing equations. No volume-fraction prefactors on any source terms.

ε_org convention. ε_org is the volume fraction of AN added per unit total solution volume — a *loading* parameter, not strictly a droplet volume fraction. Below the solubility threshold ε_sat it represents AN fully dissolved in a single aqueous phase; above ε_sat it represents AN in excess of the saturation limit, forming organic droplets. **AN concentrations use Convention A — moles per total solution volume** (see §7.2).

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1. Physical Domain

The model solves steady-state species transport across a stagnant Nernst diffusion layer of thickness δ [m] adjacent to a planar cadmium cathode (at x = 0), with the well-mixed bulk electrolyte at x = δ. Framework: Corpus et al. (Joule 2023) and Weng, Bell & Weber (PCCP 2018).

The diffusion layer contains dispersed organic droplets characterised by their volume fraction ε_org [—]. Under the local-equilibrium assumption, the organic and aqueous phases are in instantaneous equilibrium everywhere — the droplets provide a parallel diffusion pathway for species that partition into the organic phase. All electrochemical reactions occur at the electrode surface and enter the model as flux boundary conditions at x = 0. H_2 desorbs immediately and is not tracked.

```
ELECTRODE (Cd)        DIFFUSION LAYER (δ)                 BULK ELECTROLYTE
    x = 0          organic droplets in local equil.              x = δ
                   |                                      |
Tafel kinetics     |  NP transport with migration         |  Dirichlet BCs
ADPN + PN + HER    |  D_i,mix(ε_org) eff. diffusivity  |  c_i = c_i,bulk
Flux BCs           |  Phosphate buffer (OH−pathway)       |
                   |  No R_PT (local equilibrium)         |
                   |  Current conservation (φ_l)          |
```

Key findings from Bloomquist et al. (CEJ 2026, 528, 172125):

- FE_ADPN increases from <50% to >80% when ε_org exceeds the solubility limit (~0.086)
- Bubble-induced convection dominates over inlet flow regime
- High selectivity maintained at j > 200 mA cm^{-2}
- ε_org explored from 0.02 to 0.30

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2. Species and Degrees of Freedom

The model tracks **10 dissolved species** — nine are independent unknowns; Na^+ is computed from electroneutrality.

Index	Species	Charge z_i	Role
1	H^+	+1	Proton
2	OH^-	−1	Hydroxide (product of all 4 cathodic reactions)
3	$H_2PO_4^-$	−1	Buffer
4	HPO_4^{2-}	−2	Buffer
5	PO_4^{3-}	−3	Buffer

Index	Species	Charge z_i	Role
6	AN (CH_2=CHCN)	0	Acrylonitrile (reactant)
7	ADPN ($NC(CH_2)_4CN$)	0	Adiponitrile (product)
8	PN (CH_3CH_2CN)	0	Propionitrile (byproduct)
9	**TCH (1,3,6-tricyanohexane)**	**0**	**Tricyanohexane (byproduct, NEW v7)**
10	Na^+	+1	From electroneutrality (not a DOF)

v7 change: TCH appended at index 9 to keep ordering stable for v6 -> v7. Na^+ shifts to index 10 conceptually but is still recovered algebraically (not a DOF), so transport-related index loops still run `1:9` (over true DOF species).

> **TBA^+ treatment:** Tetrabutylammonium (0.02 M = 20 mol m^{-3}) is present as a selectivity agent but is lumped into the Na^+ background — both carry z = +1, and since Na^+ is recovered from electroneutrality rather than transported, no mobility mismatch enters the model. The effective background Na^+ input is c_Na,input = 3 × 500 + 20 = **1520 mol m^{-3}** (see §6.4).

Na^+ is determined algebraically from local electroneutrality:

```
c_Na = c_OH + c_H₂PO₄ + 2·c_HPO₄ + 3·c_PO₄ − c_H
```

Each finite-volume cell carries **10 degrees of freedom** (was 9 in v6): 9 log-concentrations (species 1–9) plus the electrolyte potential ϕ_l [V]. Log-concentrations are defined as u_i = ln(c_i) and recovered via c_i = exp(clamp(u_i, –50, 50)). With N_mesh = 100: **1000 total unknowns** (was 900 in v6).

2.1 DOF index map (updated v7)

Quantity	v6	**v7**
n_species (DOFs)	8	**9**
DOFs per cell (species + ϕ_l)	9	**10**
Total DOFs (N_mesh = 100)	900	**1000**
Species index k -> DOF index	`9·(ix−1) + k`	**`10·(ix−1) + k`**
ϕ_l index	`9·ix`	**`10·ix`**
Banded Jacobian halfbw	17	**19**
Banded FD colors	35	**39**

Bandwidth derivation: for cell-major ordering with `n_dof_per_cell` DOFs per cell and a 3-point stencil (current cell + nearest neighbours), `halfbw = 2·n_dof_per_cell − 1`. With `n_dof_per_cell = 10`, `halfbw = 19` and `n_colors = 2·halfbw + 1 = 39`.

2.2 TCH species inputs (NEW v7)

TCH is the trimer product $\text{3 AN + 2 H}_2\text{O + 2 e}^- \text{ -> TCH + 2 OH}^-$ (n_e_TCH = 2 per Bloomquist SI). Atom balance with n_e = 2 implies TCH = $C_9H_{11}N_3$ (one degree of unsaturation retained from 3 AN's three C=C bonds).

Quantity	Symbol	Default for v7	Source
Charge	z_TCH	0	Neutral organic
Bulk concentration	c_TCH,bulk	0 mol/m³	Reaction product, not feedstock
Molecular weight	M_TCH	161.20 g/mol	$C_9H_{11}N_3$; *<TODO: confirm vs Bloomquist FE_TCH derivation>*
D_TCH in aqueous phase	D_TCH,aq	7.0×10^{-10} m²/s	Wilke-Chang from ADPN's 9.0×10^{-10} at M=108: $D \propto M^{-0.6}$ *<TODO: confirm>*
D_TCH in organic phase	D_TCH,org	1.2×10^{-9} m²/s	Typical 1.5–2× aq for the same solute *<TODO: confirm>*
Partition coefficient	m_TCH = c_org/c_aq	1.5	TCH more lipophilic than ADPN (m_ADPN = 0.4) *<TODO: lab>* — single highest-priority unknown

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3. Governing Equations

3.1 Nernst–Planck Transport with Migration

For each species i at steady state:

$$0 = \frac{\partial}{\partial x}\left[D_{i,\text{mix}}\frac{\partial c_i}{\partial x} + \frac{z_i F}{RT} D_{i,\text{mix}}\, c_i \frac{\partial \phi_\ell}{\partial x}\right] + R_{\text{buf},i}$$

`0 = d/dx [D_i,mix × dc_i/dx + (z_i·F)/(R·T) × D_i,mix × c_i × dφ_l/dx] + R_buf,i`

Symbol	Definition	Unit
c_i	Aqueous-phase concentration of species i	mol m^{-3}
D_i,mix	Effective diffusivity through emulsion (§4, regime-aware)	m² s^{-1}
z_i	Charge number	—

Symbol	Definition	Unit
F	Faraday constant = 96,485.332	C mol^{-1}
R	Gas constant = 8.314463	J mol^{-1} K^{-1}
T	Temperature = 298.15 K (25 °C)	K
ϕ_l	Electrolyte potential (solved DOF)	V
R_buf,i	Buffer reaction source for species i (§6)	mol m^{-3} s^{-1}
x	Spatial coordinate (0 = electrode, δ = bulk)	m
δ	Diffusion layer thickness	m

No R_PT term. Under the local-equilibrium assumption, the organic and aqueous phases equilibrate instantaneously and there is no driving force for interphase transfer. All multiphase effects are captured through D_i,mix.

No prefactor on R_buf. Buffer reactions occur in the aqueous phase, the rate law uses aqueous concentrations, and c_i is an aqueous concentration. No correction is needed.

3.2 Current Conservation (Potential Equation)

The 9th equation per cell enforces zero net ionic current in the diffusion layer. No external current flows through the solution between x = 0 and x = δ — current enters and exits only at the electrode. This constraint determines the electrolyte potential ϕ_l at every position:

$$\frac{\partial}{\partial x}\left[\sum_i z_i N_i\right] = 0$$

where N_i is the Nernst–Planck flux of species i and the sum runs over charged species only. In the FV discretisation, this becomes: the sum of ionic SG fluxes at the right face of each cell must equal the sum at the left face.

> **Well-posedness (gauge fix):** The current-conservation equation `d/dx[Σ z_i N_i] = 0` is invariant under a uniform shift of ϕ_l — the system has one null direction. The Dirichlet condition ϕ_l(δ) = 0 (§7.2) kills this null direction and makes the Jacobian non-singular. Without it, Newton cannot converge.

3.3 Scharfetter–Gummel Flux Discretisation

Standard centred-difference flux approximations become numerically unstable when the dimensionless migration Péclet number α = z_i F Δϕ/(RT) exceeds ~2 per cell, producing spurious oscillations. The SG scheme uses an exponential fitting to handle arbitrarily large α stably.

$$J_i = -\frac{D_{i,\text{mix}}}{\Delta x}\left[B(\alpha)\, c_R - B(-\alpha)\, c_L\right], \quad \alpha = \frac{z_i F(\phi_R - \phi_L)}{RT}$$

with the Bernoulli functions:

$$B(\alpha) = \frac{\alpha}{e^{\alpha} - 1}, \qquad B(-\alpha) = \frac{\alpha\, e^{\alpha}}{e^{\alpha} - 1}$$

where c_L, c_R are aqueous concentrations in the left and right cells [mol m^{-3}]; ϕ_L, ϕ_R are electrolyte potentials [V]; and Δx is the distance between cell centres [m]. α is clamped to [–700, 700] to avoid floating-point overflow.

> **Taylor expansion for small α (required for AD).** At α = 0 exactly (e.g. the equilibrium initial guess), the naïve Bernoulli form divides by zero. A hard `if abs(α) < 1e-10: return centered_diff` branch *also* fails for automatic differentiation (AD) — it creates a spurious zero derivative at the α = 0 seed point because the centered-difference form has no ϕ dependence. **Use the Taylor series for |α| < 0.01** to keep the function smooth *and* its ϕ-derivative continuous through α = 0:
>
> ```
> B(α)  = 1 − α/2 + α²/12 − α⁴/720 + ...    (for |α| < 0.01)
> B(−α) = 1 + α/2 + α²/12 + α⁴/720 + ...
> ```

```julia
function sg_flux(c_L, c_R, phi_L, phi_R, D_mix, z_i, dx, T=298.15)
    z_i == 0 && return -D_mix * (c_R - c_L) / dx
    alpha = clamp(z_i * F * (phi_R - phi_L) / (R * T), -700.0, 700.0)
    if abs(alpha) < 0.01
        a2 = alpha * alpha
        B_pos = 1.0 - alpha / 2.0 + a2 / 12.0
        B_neg = 1.0 + alpha / 2.0 + a2 / 12.0
    else
        ea  = exp(alpha)
        den = ea - 1.0
        B_pos = alpha / den
        B_neg = alpha * ea / den
    end
    return -D_mix / dx * (B_pos * c_R - B_neg * c_L)
end
```

---

4. Mixture-Averaged Diffusivities — Regime-Aware

4.1 Two regimes

AN loading partitions the electrolyte into two regimes:

Regime	Threshold	Physical state	D_i,mix
Single-phase	ε_org < ε_sat	All AN dissolved in aqueous phase — no droplets	**D_i,mix = D_i,aq** (no mixing)

Regime	Threshold	Physical state	D_i,mix
Two-phase	ε_org >= ε_sat	Aqueous phase saturated at C_AN_SAT; excess AN forms organic droplets	**D_i,mix = ε_org · D_i,org + (1 – ε_org) · D_i,aq** (arithmetic mean)

The threshold ε_sat is determined by the aqueous saturation concentration of AN (§7.2):

$$\varepsilon_{\text{sat}} = \frac{C_{\text{AN,SAT}}}{\rho_{\text{AN}}/M_{\text{AN}}} = \frac{1310}{15{,}191} \approx \mathbf{0.0862}$$

Below ε_sat there are no organic droplets at all; the "organic diffusivity pathway" is physically absent. Only above ε_sat does the volume-averaged diffusivity become meaningful.

Ions are insoluble in the organic phase, so D_i,org = 0 for all ions. The two-phase formula becomes:

$$D_{i,\text{mix}}^{\text{ion}} = (1 - \varepsilon_{\text{org}})\, D_{i,\text{aq}} \quad (\varepsilon_{\text{org}} \geq \varepsilon_{\text{sat}})$$

Increasing ε_org above ε_sat **reduces ionic transport** (ohmic penalty) while **enhancing neutral-species transport** (AN parallel pathway). Neutral organic diffusivities are estimated via the Stokes–Einstein viscosity ratio: D_i,org ~= D_i,aq × (μ_aq/μ_org) ~= 2.6 × D_i,aq (μ_H_2O = 0.89 mPa·s, μ_AN = 0.34 mPa·s at 25°C).

Species	D_aq (m²/s)	D_org (m²/s)	Notes
AN	2.30×10^{-9}	6.00×10^{-9}	Stokes–Einstein ×2.6
ADPN	1.50×10^{-9}	3.90×10^{-9}	Stokes–Einstein ×2.6
PN	2.30×10^{-9}	6.00×10^{-9}	Stokes–Einstein ×2.6
TCH	**0.70×10^{-9}**	**1.20×10^{-9}**	**Wilke-Chang scaling from ADPN at M=161 (NEW v7)**
H^+	9.31×10^{-9}	0	Insoluble in organic
OH^-	5.27×10^{-9}	0	Insoluble in organic

Species	D_aq (m²/s)	D_org (m²/s)	Notes
$H_2PO_4^-$	0.846×10^{-9}	0	Insoluble in organic
HPO_4^{2-}	0.690×10^{-9}	0	Insoluble in organic
PO_4^{3-}	0.610×10^{-9}	0	Insoluble in organic
Na^+	1.33×10^{-9}	0	Insoluble in organic

D_aq from Suwanvaipattana et al. (2017) and CRC Handbook. All values at T = 25 °C. TCH defaults are estimates pending Bui Lab confirmation.

```
const T    = 298.15   # K, operating temperature

# v7: arrays grow from length 8 to length 9 (TCH appended)
const D_aq = [9.31e-9, 5.27e-9, 0.846e-9, 0.69e-9, 0.61e-9,   # H⁺ OH⁻ H₂PO₄⁻
 ↪  HPO₄²⁻ PO₄³⁻
             2.3e-9, 1.5e-9, 2.3e-9, 0.70e-9]                # AN ADPN PN TCH (NEW)
const D_org = [0.0, 0.0, 0.0, 0.0, 0.0,                        # ions: insoluble
              6.0e-9, 3.9e-9, 6.0e-9, 1.20e-9]               # AN ADPN PN TCH (NEW)

const EPS_ORG_SAT = C_AN_SAT / MOLAR_DENSITY_AN    # ~= 0.0862

function D_mix(i, eps_org)
    if eps_org < EPS_ORG_SAT
        return D_aq[i]                                 # single-phase: no organic
    else
        return eps_org * D_org[i] + (1.0 - eps_org) * D_aq[i]   # two-phase:
         ↪  arithmetic
    end
end
```

Effect of ε_org on Mixture Diffusivities

ε_org	Regime	D_AN,mix (m²/s)	D_OH,mix (m²/s)
0.02	single-phase	2.30×10^{-9}	5.27×10^{-9}
0.05	single-phase	2.30×10^{-9}	5.27×10^{-9}
0.08	single-phase	2.30×10^{-9}	5.27×10^{-9}
0.0862	**threshold**	$\mathbf{2.62 \times 10^{-9}}$	$\mathbf{4.82 \times 10^{-9}}$
0.09	two-phase	2.63×10^{-9}	4.80×10^{-9}
0.15	two-phase	2.86×10^{-9}	4.48×10^{-9}
0.25	two-phase	3.23×10^{-9}	3.95×10^{-9}

ε_org	Regime	D_AN,mix (m²/s)	D_OH,mix (m²/s)
0.30	two-phase	3.41×10^{-9}	3.69×10^{-9}

The transition at ε_sat is a ~14% step in D_AN (jumping from pure D_aq to the arithmetic mean). This reflects the physical onset of organic droplets providing a parallel transport pathway.

AN transport improves by 48% at ε_org = 0.30 (relative to single-phase D_aq) while OH^- transport degrades by 30% — this quantifies the selectivity–conductivity trade-off in the two-phase regime.

4.2 m_i-Corrected Upgrade (Future)

If the arithmetic mean is too weak to reproduce the experimental FE enhancement, the physically correct local-equilibrium effective diffusivity includes the partition coefficient m_i = c_i,org/c_i,aq:

$$D_{i,\text{eff}} = (1 - \varepsilon_{\text{org}})\, D_{i,\text{aq}} + \varepsilon_{\text{org}}\, m_i\, D_{i,\text{org}} \quad (\varepsilon_{\text{org}} \geq \varepsilon_{\text{sat}})$$

For AN (m_AN = 11.59) at ε_org = 0.25: D_AN,eff = 19.1×10^{-9} m²/s vs D_AN,mix = 3.23×10^{-9}.
For ions: m_i = 0, so D_ion,eff = (1−ε_org) D_i,aq — no change.

```
# One-line switch from arithmetic to m_i-corrected (two-phase only):
D_mix(i, eo) = eo < EPS_ORG_SAT ? D_aq[i] : eo * D_org[i] + (1-eo) * D_aq[i]
↪  # arithmetic
D_eff(i, eo) = eo < EPS_ORG_SAT ? D_aq[i] : eo * m[i] * D_org[i] + (1-eo) * D_aq[i]
↪  # m-corrected
```

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5. Electrochemical Kinetics

5.1 Competing Cathodic Reactions

Reaction	Stoichiometry	n_e
R1: Electrohydrodimerization (ADPN)	2 AN + 2 H_2O + 2 e^- -> ADPN + 2 OH^-	2
R2: Hydrogenation (PN)	AN + 2 H_2O + 2 e^- -> PN + 2 OH^-	2
R3: Hydrogen evolution (HER)	2 H_2O + 2 e^- -> H_2 + 2 OH^-	2
R4: Trimerization (TCH, NEW v7)	**3 AN + 2 H_2O + 2 e^- -> TCH + 2 OH^-**	**2**

All four reactions transfer n_e = 2 electrons per product molecule and produce 2 OH^- per 2 electrons (= 1 OH^- per electron — same convention across all reactions). The n_e_TCH = 2 stoichiometry is taken from the Bloomquist SI; the resulting $C_9H_{11}N_3$ TCH formula reflects retention of one degree of unsaturation from the three AN C=C bonds.

5.2 Tafel Rate Expressions (updated v7)

All reactions operate far from equilibrium; the cathodic Tafel approximation applies. The current densities j_r [A m^{-2}] at the electrode surface are:

R1 — ADPN (AN reaction order n_ADN, fit parameter):

$$j_1 = j_{0,1}\left(\frac{c_{\text{AN}}}{c_{\text{ref}}}\right)^{n_{\text{ADN}}} \text{exp}\left(-\frac{\alpha_{c,1} F\,\eta_1}{RT}\right)$$

R2 — PN (AN reaction order n_PN, fit parameter):

$$j_2 = j_{0,2}\left(\frac{c_{\text{AN}}}{c_{\text{ref}}}\right)^{n_{\text{PN}}} \text{exp}\left(-\frac{\alpha_{c,2} F\,\eta_2}{RT}\right)$$

R3 — HER (independent of AN):

$$j_3 = j_{0,3}\,\text{exp}\left(-\frac{\alpha_{c,3} F\,\eta_3}{RT}\right)$$

R4 — TCH (AN reaction order n_TCH, fit parameter; NEW v7):

$$j_4 = j_{0,\text{TCH}}\left(\frac{c_{\text{AN}}}{c_{\text{ref}}}\right)^{n_{\text{TCH}}} \text{exp}\left(-\frac{\alpha_{c,\text{TCH}} F\,\eta_4}{RT}\right)$$

The overpotential for each reaction: η_r = (ϕ_s – ϕ_l) – E°_r. v7 keeps E°_1 = E°_2 = E°_TCH = –1.30 V vs SHE *<TODO: confirm E°_TCH against Bui Lab data>*.

Symbol	Definition	Unit
j_r	Current density for reaction r (positive = cathodic)	A m^{-2}
j_0,r	Exchange current density for reaction r	A m^{-2}
c_AN	Aqueous AN concentration at the electrode surface	mol m^{-3}
c_ref	Reference concentration = 1,000 mol m^{-3} (= 1 M)	mol m^{-3}
α_c,r	Cathodic transfer coefficient for reaction r	—
n_r	**AN reaction order for reaction r (fit param in v7 for r in {ADN, PN, TCH})**	**—**
η_r	Overpotential for reaction r	V
ϕ_s	Applied cathode potential	V vs SHE

Symbol	Definition	Unit
ϕ_l	Solved electrolyte potential at the electrode	V
E°_r	Standard reduction potential for reaction r	V vs SHE

v6 -> v7 kinetics change. v6 hardcoded n_ADN = 2 and n_PN = 1 (matching collision-theory molecularity). v7 promotes both — and adds n_TCH (initial 3) — to fit parameters via the `KIN_OVERRIDE` Ref in `kinetics.jl`. Default fallback (no override) reads `n_ADN = 2, n_PN = 1, n_TCH = 3` from Params, so Stages 1 / 2 / 2m / 3 stay byte-identical to v5/v6 in their default-kinetics paths.

Unit note: j_0,r is given in the fitting table (§9.2) in mA cm^{-2}. Convert to SI via ×10: 1 mA cm^{-2} = 10 A m^{-2}. c_ref = 1000 mol m^{-3} so that (c_AN/c_ref) is dimensionless; c_AN can reach C_AN_SAT ~= 1310 mol m^{-3} at saturation, which is physical.

The exponent on `(c_AN/c_ref)` is the central selectivity feature: at high j the surface c_AN drops, and the reaction with the largest n suffers the steepest falloff. Empirically (per the v6.x and v7 fits) the AN orders sit *below* their textbook molecularity values — an indication that Langmuir-Hinshelwood surface coverage saturation (rather than collision theory) governs the practical kinetics.

The Faradaic efficiency for product r in {ADPN, PN, TCH}:

$$\text{FE}_r = \frac{j_r}{j_1 + j_2 + j_3 + j_4} \times 100\%$$

6. Phosphate Buffer Chemistry — OH^--Pathway

6.1 Equilibria

Three equilibria are tracked. H_3PO_4 dissociation (pK = 2.12) is omitted because it is negligible at the operating pH of 12–13. Water autoprotolysis uses the classical Eigen–De Maeyer rate; phosphate deprotonations use the **OH^--pathway** forms, which are the physically dominant kinetic paths at high pH.

#	Reaction (forward direction)	K_eq (25°C)
R1	$H_2O \Leftrightarrow H^+ + OH^-$	K_w = 10^{-14} M^2 = 10^{-8} $(mol/m^3)^2$
R2	**$OH^- + H_2PO_4^- \Leftrightarrow HPO_4^{2-} + H_2O$**	K_eq,R2 = Ka2/K_w = 6.3 × 10^6 M^{-1} = 6.3 × 10^3 m^3/mol
R3	**$OH^- + HPO_4^{2-} \Leftrightarrow PO_4^{3-} + H_2O$**	K_eq,R3 = Ka3/K_w = 45 M^{-1} = 4.5 × 10^{-2} m^3/mol

Equivalent acid-dissociation constants (consistent with the OH^--pathway K_eq values):

Symbol	Value	Unit
K_w	10^{-8}	$(mol/m^3)^2$
Ka2	K_eq,R2 · K_w = 6.3×10^{-5}	mol/m^3 (pKa2 = 7.20)
Ka3	K_eq,R3 · K_w = 4.5×10^{-10}	mol/m^3 (pKa3 = 12.35)

Why OH^--pathway, not H^+-pathway? At pH ~= 13, c_H ~= 10^{-10} mol/m^3 and c_OH ~= 10^2 mol/m^3. The classical form `H₂PO₄⁻ <=> H⁺ + HPO₄²⁻` requires a reverse rate `k_r · c_H · c_HPO₄` that offsets the forward rate through a tiny c_H — driving k_r to unphysically huge values (~10^{10} $m^3/mol/s$). The OH^--pathway `OH⁻ + H₂PO₄⁻ -> HPO₄²⁻ + H₂O` keeps c_OH (large) in the forward rate and gives physically reasonable rate constants. Both give identical thermodynamic equilibrium.

Reverse rate constants are derived from K_eq: **k_r = k_f / K_eq** (never set independently).

6.2 Rate Expressions

For each equilibrium, the net rate [mol m^{-3} s^{-1}] is forward minus reverse:

$$r_1 = k_{1,f} - k_{1,r}\, c_{\mathrm{H^+}}\, c_{\mathrm{OH^-}}$$

$$r_2 = k_{2,f}\, c_{\mathrm{OH^-}}\, c_{\mathrm{H_2PO_4^-}} - k_{2,r}\, c_{\mathrm{HPO_4^{2-}}}$$

$$r_3 = k_{3,f}\, c_{\mathrm{OH^-}}\, c_{\mathrm{HPO_4^{2-}}} - k_{3,r}\, c_{\mathrm{PO_4^{3-}}}$$

Species allocation (OH^--pathway stoichiometry):

$$R_{\mathrm{buf}}[\mathrm{H^+}] = r_1 \qquad R_{\mathrm{buf}}[\mathrm{OH^-}] = r_1 - r_2 - r_3$$
$$R_{\mathrm{buf}}[\mathrm{H_2PO_4^-}] = -r_2 \qquad R_{\mathrm{buf}}[\mathrm{HPO_4^{2-}}] = r_2 - r_3 \qquad R_{\mathrm{buf}}[\mathrm{PO_4^{3-}}] = r_3$$

R_buf = 0 for AN, ADPN, and PN. **No prefactor** — c_i is aqueous; R_buf is rate per aqueous volume. Charge conservation Σ z_i · S_i = 0 holds for each individual reaction.

```
function buffer_sources!(R, c_H, c_OH, c_H2PO4, c_HPO4, c_PO4, alpha_buf=1.0)
    r1 = k1f - k1r * c_H * c_OH                               # H₂O <=> H⁺ + OH⁻
    r2 = k2f * c_OH * c_H2PO4 - k2r * c_HPO4                  # OH⁻ + H₂PO₄⁻ <=>
↪   HPO₄²⁻ + H₂O
    r3 = k3f * c_OH * c_HPO4  - k3r * c_PO4                   # OH⁻ + HPO₄²⁻ <=> PO₄³⁻
↪   + H₂O
    R[1] = alpha_buf * r1                          # H⁺     — water only
    R[2] = alpha_buf * (r1 - r2 - r3)              # OH⁻    — water + consumed by
↪   phosphates
```

```
    R[3] = alpha_buf * (-r2)                   # H₂PO₄⁻
    R[4] = alpha_buf * (r2 - r3)              # HPO₄²⁻
    R[5] = alpha_buf * r3                     # PO₄³⁻
    R[6:8] .= 0.0                              # AN, ADPN, PN
end
```

The optional `alpha_buf` argument (default 1.0) is the buffer continuation parameter used during bootstrap (see §10.6, §12).

6.3 Rate Constants (SI units)

Constant	Value	Unit	Source / Derivation
k_1,f	**1.4**	mol/(m³·s)	Eigen–De Maeyer 1955; 1.4 × 10^{-3} M/s × 1000
k_1,r	**1.4 × 10^8**	m³/(mol·s)	= k_1,f / K_w (== 1.4 × 10^{11} M^{-1} s^{-1})
k_2,f	**1 × 10^5**	m³/(mol·s)	10^8 M^{-1} s^{-1} / 1000 (second-order forward)
k_2,r	**~= 15.87**	s^{-1}	= k_2,f / K_eq,R2 (unimolecular reverse)
k_3,f	**2 × 10^3**	m³/(mol·s)	2 × 10^6 M^{-1} s^{-1} / 1000
k_3,r	**~= 4.44 × 10^4**	s^{-1}	= k_3,f / K_eq,R3

Dimensional consistency check: `mol/(m³·s) − m³/(mol·s) · (mol/m³) · (mol/m³)` = `mol/(m³·s)` for R1; `m³/(mol·s) · (mol/m³)² − s⁻¹ · (mol/m³)` = `mol/(m³·s)` for R2, R3. [OK]

6.4 Bulk Equilibrium Initial Guess

This is the most critical step for convergence. Buffer reactions are stiff: forward and reverse rates are both large but nearly cancel at equilibrium. An initial guess that is even slightly off-equilibrium produces buffer source terms of O(10^3–10^5) mol m^{-3} s^{-1} — orders of magnitude larger than transport fluxes — causing Newton to fail or stall even at α_buf = 0.001.

The fix: solve the phosphate charge balance exactly before starting Newton.

The equilibrium speciation follows from two constraints:

Phosphate conservation:

$$c_{\text{H}_2\text{PO}_4} + c_{\text{HPO}_4} + c_{\text{PO}_4} = C_{P,\text{total}}$$

Expressing each species in terms of c_H alone:

$$c_{\text{HPO}_4} = \frac{C_{P,\text{total}}}{c_H/K_{a2} + 1 + K_{a3}/c_H}, \quad c_{\text{H}_2\text{PO}_4} = \frac{c_H\, c_{\text{HPO}_4}}{K_{a2}}, \quad c_{\text{PO}_4} = \frac{K_{a3}\, c_{\text{HPO}_4}}{c_H}$$

Charge balance (one equation in c_H):

$$c_{\text{Na,input}} + c_H = \frac{K_w}{c_H} + c_{\text{H}_2\text{PO}_4} + 2\, c_{\text{HPO}_4} + 3\, c_{\text{PO}_4}$$

where c_Na,input = 3 × C_Na_3PO_4 + C_TBA-OH = 3 × 500 + 20 = **1520 mol m^{-3}** (TBA^+ lumped into Na^+, see §2).

Solve via 1D bisection on $\log_{10}$(c_H) over pH in [6, 15]:

```
const K_w  = 1.0e-8     # (mol/m³)², water autoprotolysis at 25°C
const K_a2 = 6.3e-5     # mol/m³, H₂PO₄⁻ <-> H⁺ + HPO₄²⁻ at 25°C
const K_a3 = 4.5e-10    # mol/m³, HPO₄²⁻ <-> H⁺ + PO₄³⁻ at 25°C

function solve_phosphate_equilibrium(C_P_total=500.0, c_Na_input=1520.0)
    function charge_balance(log10_cH)
        cH    = 10.0^log10_cH
        cOH   = K_w / cH
        denom = cH / K_a2 + 1.0 + K_a3 / cH
        cHPO4  = C_P_total / denom
        cH2PO4 = cH * cHPO4 / K_a2
        cPO4   = K_a3 * cHPO4 / cH
        return c_Na_input + cH - (cOH + cH2PO4 + 2.0*cHPO4 + 3.0*cPO4)
    end
    log10_cH_eq = bisect(charge_balance, -12.0, -3.0)   # or Roots.find_zero
    cH = 10.0^log10_cH_eq
    ...
    return (H=cH, OH=cOH, H2PO4=cH2PO4, HPO4=cHPO4, PO4=cPO4,
            Na=c_Na_input, pH=-log10(cH/1000.0))
end
```

Verification — always run before starting Newton:

```
c_eq = solve_phosphate_equilibrium()
@assert abs(c_eq.H * c_eq.OH                - K_w)  / K_w  < 1e-8   "K_w mismatch"
@assert abs(c_eq.H * c_eq.HPO4 / c_eq.H2PO4 - K_a2) / K_a2 < 1e-8   "Ka2 mismatch"
@assert abs(c_eq.H * c_eq.PO4  / c_eq.HPO4  - K_a3) / K_a3 < 1e-8   "Ka3 mismatch"

R_check = zeros(8)
buffer_sources!(R_check, c_eq.H, c_eq.OH, c_eq.H2PO4, c_eq.HPO4, c_eq.PO4)
@assert maximum(abs.(R_check)) < 1e-6  "Buffer not at equilibrium"
```

Expected equilibrium values for 0.5 M Na_3PO_4 + 0.02 M TBA-OH at 25°C with OH^--pathway constants:

Species	Value	Unit
c_H	~= 9.4×10^{-11}	mol m^{-3}
c_OH	~= 106	mol m^{-3}
c_H_2PO_4	~= 1.3×10^{-4}	mol m^{-3}
c_HPO_4	~= 86	mol m^{-3}
c_PO_4	~= 414	mol m^{-3}
c_Na	1520	mol m^{-3}
pH	~= 13.03	—

Most phosphate sits as PO_4^{3-} since pH ~= 13.03 is well above pKa3 = 12.35.

Building the full DOF initial guess vector:

```
function make_initial_guess(N_mesh, c_eq, eps_org; c_seed=1e-3)
    u0 = zeros(9 * N_mesh)
    log_c0 = [log(c_eq.H),    log(c_eq.OH),
              log(c_eq.H2PO4), log(c_eq.HPO4), log(c_eq.PO4),
              log(c_AN_bulk(eps_org)),      # see §7.2 — requires ε_org > 0
              log(c_seed),                   # ADPN: small seed
              log(c_seed)]                   # PN:   small seed
    for ix in 1:N_mesh
        u0[9*(ix-1)+1 : 9*(ix-1)+8] .= log_c0
        u0[9*(ix-1)+9] = 0.0
    end
    return u0
end
```

With this initialisation and α_buf = 0 (buffer off), α_kin = 0 (kinetics off), the residual is **identically zero**: no concentration gradients, no current, no buffer sources. Newton converges in 0 iterations — the ideal starting point for the bootstrap protocol (§12). Requires ε_org > 0 so that c_AN,bulk > 0 (the ε_org = 0 limit is now truly pathological — see §7.2).

---

7. Boundary Conditions

7.1 Electrode (x = 0): Faradaic Flux (updated v7)

The species fluxes at the electrode surface are set by Faraday's law applied to the four cathodic reactions. Two AN molecules are consumed per ADPN produced, one AN per PN, and **three AN per TCH (NEW v7)**:

$$N_{\mathrm{AN}}\big|_{x=0} = -\frac{2j_1 + j_2 + 3j_4}{2F}$$

$$N_{\mathrm{ADPN}}\big|_{x=0} = +\frac{j_1}{2F} \qquad N_{\mathrm{PN}}\big|_{x=0} = +\frac{j_2}{2F} \qquad N_{\mathrm{TCH}}\big|_{x=0} = +\frac{j_4}{2F}$$

$$N_{\mathrm{OH^-}}\big|_{x=0} = +\frac{j_1 + j_2 + j_3 + j_4}{F}$$

All other species are non-electroactive: N_i(0) = 0 for H^+ and phosphates.

> **v7 derivation check.** Each reaction r contributes a product formation rate `j_r/(n_e_r · F)` [mol/m²/s] at the electrode (n_e = 2 for all four reactions in v7). Reactant-AN consumption follows from molecularity: ADPN consumes 2 AN per molecule -> 2 · `j_1/(2F)` = `j_1/F`; PN consumes 1 AN -> `j_2/(2F)`; TCH consumes 3 AN -> 3 · `j_4/(2F)`. Sum and negate -> –(2 j_1 + j_2 + 3 j_4)/(2F).

The electrolyte potential ϕ_l(0) is **not prescribed** — it is determined by the current conservation equation (§3.2). The overpotentials in the Tafel rate expressions use this solved value.

7.2 Bulk (x = δ): Dirichlet — AN via Convention A

At the bulk boundary, all species are fixed at their well-mixed bulk values and the electrolyte potential is referenced to zero:

$$c_i(\delta) = c_{i,\text{bulk}} \qquad \phi_\ell(\delta) = 0$$

The bulk equilibrium concentrations of H^+, OH^-, and the three phosphate species are computed by `solve_phosphate_equilibrium()` (§6.4). ADPN and PN bulk concentrations are set to small seed values (~10^{-3} mol m^{-3}) to avoid log(0) in the initial guess.

AN bulk concentration (Convention A — moles per total solution volume):

AN is loaded at a volume fraction ε_org relative to the total solution volume. With ρ_AN = 806 kg/m³ (neat AN density), M_AN = 53.06 g/mol, and m_AN = 11.59 (partition coefficient), the molar density of neat AN is ρ_AN/M_AN = 15,191 mol/m³, and the aqueous saturation concentration is C_AN_SAT = ρ_AN/(M_AN·m_AN) = **1,310 mol/m³**.

Regime	Condition	c_AN,bulk	Physical picture
Single-phase	ε_org < ε_sat ~= 0.0862	**ε_org · ρ_AN/M_AN**	All AN dissolved; c_AN linearly rises with loading
Two-phase	ε_org >= ε_sat	**C_AN_SAT = 1310 mol/m³**	Aqueous saturated; excess AN in droplets

Saturation threshold: ε_sat · (ρ_AN/M_AN) = C_AN_SAT => **ε_sat = 1310 / 15 191 ~= 0.0862**. Above ε_sat, increasing ε_org does **not** increase c_AN,bulk — it is fixed at 1310 mol/m³. The effect of higher ε_org enters entirely through D_i,mix (§4): organic droplets provide a parallel transport pathway for AN.

```
const M_AN              = 0.05306       # kg/mol
const RHO_AN            = 806.0         # kg/m³
const m_AN              = 11.59         # partition coefficient
const MOLAR_DENSITY_AN = RHO_AN / M_AN # ~= 15 191 mol/m³
const C_AN_SAT          = MOLAR_DENSITY_AN / m_AN       # ~= 1310 mol/m³
const EPS_ORG_SAT       = C_AN_SAT / MOLAR_DENSITY_AN   # ~= 0.0862

function c_AN_bulk(eps_org)
    eps_org < EPS_ORG_SAT ? eps_org * MOLAR_DENSITY_AN : C_AN_SAT
end
```

> **ε_org = 0 is non-physical.** It represents zero AN in the system — no ADPN, no PN, no kinetic activity. All sweeps must start at ε_org > 0. The Stage 1 reference point is `ε_org = 0.02` (c_AN,bulk ~= 304 mol/m³; single-phase).

Numerical table at the planned sweep points:

ε_org	c_AN,bulk [mol/m³]	D_AN,mix [m²/s]	Regime
0.02	304	2.30×10^{-9}	single-phase
0.05	760	2.30×10^{-9}	single-phase
0.08	1 215	2.30×10^{-9}	single-phase
0.0862	**1 310** (= C_AN_SAT)	2.62×10^{-9}	threshold
0.09	1 310	2.63×10^{-9}	two-phase
0.15	1 310	2.86×10^{-9}	two-phase
0.25	1 310	3.23×10^{-9}	two-phase
0.30	1 310	3.41×10^{-9}	two-phase

---

8. Three-Parameter Sweep

Parameter	Range	Points
V (cathode potential)	−1.0 to −2.5 V vs SHE	Newton continuation
δ (boundary layer thickness)	10, 20, 50, 100, 200 μm	5
ε_org (AN loading)	**0.02, 0.05, 0.08, 0.15, 0.25, 0.30**	6

The ε_org sweep spans both regimes: {0.02, 0.05, 0.08} single-phase (below ε_sat ~= 0.0862), {0.15, 0.25, 0.30} two-phase. The transition at ε_sat is directly observable. ε_org = 0 is never swept (non-physical).

Total: 30 Newton continuation sweeps -> 3D performance map FE_ADPN(V, δ, ε_org).

The diffusion-only limiting current is:

$$j_{\text{lim}} = \frac{n_e \, F \, D_{\text{AN,mix}} \, c_{\text{AN,bulk}}}{\delta}$$

Under the local-equilibrium assumption with the arithmetic-mean D_mix (two-phase regime), the enhancement at higher ε_org comes through the D_AN,mix > D_AN,aq channel. With the m_i-corrected D_eff (§4.2), the enhancement would be much larger. Comparing model predictions under both assumptions against Bloomquist data will determine which is correct.

---

9. Parameter Tables

9.1 Physical Constants

Symbol	Value	Unit
F	96,485.332	C mol^{-1}
R	8.314463	J mol^{-1} K^{-1}
T	298.15	K (25 °C)

9.2 Kinetic Fitting Parameters (extended v7)

Parameter	Initial	Range	Unit	Basis
E^0_1 (ADPN)	−1.3	−1.0 to −1.5	V vs SHE	Onset (Mathison JACS 2025)
E^0_2 (PN)	−1.3	−1.0 to −1.5	V vs SHE	Same reaction centre
E^0_3 (HER)	−0.83	Fixed	V vs SHE	Nernst at pH 14
E°_TCH (NEW)	**−1.3** *<TODO: confirm>*	(fixed)	V vs SHE	Defaults to ADPN onset until Bui Lab provides TCH-specific value
$j_{0,1}$	10^{-4}	10^{-6} to 10^{-2}	mA cm^{-2}	Irreversible organic reduction
$j_{0,2}$	10^{-4}	10^{-6} to 10^{-2}	mA cm^{-2}	Similar to R1
$j_{0,3}$ (Cd)	**2.666×10^{-6}**	**(frozen v7)**	mA cm^{-2}	**v6.x converged value; HER no longer fit in v7**
$j_{0,TCH}$ (NEW)	**10^{-4}**	**10^{-6} to 10^{-2}**	**mA cm^{-2}**	**Initial fit guess; matches $j_{0,1}$ / $j_{0,2}$ v5 defaults**
α_c,1	0.5	0.3–0.7	—	Single-electron RDS
α_c,2	0.5	0.3–0.7	—	Single-electron RDS
α_c,3	**0.390**	**(frozen v7)**	—	**v6.x converged value**
α_c,TCH (NEW)	**0.5**	**0.30–0.70**	**—**	**Initial fit guess**

Parameter	Initial	Range	Unit	Basis
n_ADN (NEW v7 fit param)	**2.0**	**1.0–3.0**	—	**Initial = collision-theory molecularity; bounds allow LH-saturation regimes**
n_PN (NEW v7 fit param)	**1.0**	**0.5–2.0**	—	**Initial = collision-theory molecularity**
n_TCH (NEW v7 fit param)	**3.0**	**1.0–3.0**	—	**Initial = trimer molecularity**

Multiply mA cm^{-2} by 10 to convert to A m^{-2} for use in the rate expressions.

> **Why HER is frozen in v7.** The v6.x first fit returned j_0,3 = 2.67×10^{-5} A/m² (= 2.666×10^{-6} mA/cm²) and α_c,3 = 0.390. Pure-Cd literature (Trasatti 1972) gives j_0,3 ~= 10^{-7} A/m² and α_c ~= 0.50, ~250× lower than the actual Bloomquist surface produces. The v6.x values reflect the *real* electrode (organic phase, bubbles, possible surface oxidation) and already balance against the FE_HER-by-difference column. Freezing them at the v6.x converged values (a) reduces fit dimension from 11 to 9 with negligible loss in predictive power and (b) avoids the optimiser chasing the noisiest column in the dataset. If v7 residuals show systematic FE_HER trend with j or ε_org, HER can be thawed back into the fit in a v8 follow-up.

9.3 Electrolyte Composition (Modestino Group)

Component	Concentration	Role
Na_3PO_4	0.5 M (500 mol m^{-3})	Buffer + supporting electrolyte
TBA-OH	0.02 M (20 mol m^{-3})	Selectivity agent — lumped into Na^+ (see §2)
EDTA disodium	0.03 M	Metal chelator — not tracked (negligible vs. phosphate)

Component	Concentration	Role
Acrylonitrile	**ε_org in [0.02, 0.30] volume fraction**	Reactant (sweep parameter; never 0)
Cathode	Cd foil	High hydrogen overpotential

9.4 Partition Coefficients (for m_i upgrade)

Species	m_i = c_org/c_aq	Source
AN	11.59	Suwanvaipattana 2017
SN	7.72	Suwanvaipattana 2017
ADPN	0.4	Suwanvaipattana 2017
TCH (NEW v7)	**1.5** *<TODO: lab>*	**Estimate; TCH more lipophilic than ADPN**
Ether	19.02	Suwanvaipattana 2017
All ions	0	Insoluble

---

10. Numerical Methods

10.1 Mesh

Cell-centred finite-volume on a geometrically graded 1D mesh with **N_mesh = 100** cells. Cells are finest at x = 0 (electrode) and coarsen toward x = δ (bulk). Parameterised by stretch factor s = dx_max/dx_min (default s = 10):

```
r = s^(1/(N−1))              (common ratio)
dx_min = δ × (r−1) / (r^N − 1)
dx[k] = dx_min × r^(k−1)      for k = 1, …, N
```

At δ = 50 μm with N = 100 and s = 10: dx_min ~= 0.13 μm, dx_max ~= 1.3 μm. SG flux for charged species; centred differences for neutrals (reduces to SG with z = 0).

10.2 Jacobian — Block-Tridiagonal Bandwidth is b = 19 (v7; was 17 in v6)

The Jacobian has **block-tridiagonal structure** with block size 10 in v7 (one block per cell's DOFs = 9 species + 1 ϕ_l). In cell-major layout, coupling of cell ix to cells ix–1, ix, ix+1 gives:

$$\text{half-bandwidth } b = 2 \cdot B - 1 = 2 \cdot 10 - 1 = \mathbf{19} \quad (\text{v7})$$

[!] **Correcting guide v4:** Earlier versions stated b = 9; that's the *block* size, not the bandwidth. With too-small b the banded-FD column-grouping aliases unrelated columns onto the same rows and produces an exactly-singular Jacobian. Use `b = 19` (v7) and `n_colors = 2b + 1 = 39` FD perturbations.

```
const JAC_BLOCK  = 10                    # v7: 9 species + 1 φ_l (was 9 in v6)
const JAC_HALFBW = 2 * JAC_BLOCK - 1    # = 19  (was 17 in v6)
```

Banded FD: 39 residual evaluations per Jacobian (vs 1000 for dense) -> ~26× speedup per step. Sparse LU via `SparseArrays.jl`.

> **v7 update note.** v6 had `JAC_BLOCK = 9`, `JAC_HALFBW = 17`, `n_colors = 35`. The 9 -> 10 species -> 10 -> 19 -> 39 step is mechanical but easy to miss in one obscure place. After updating `assembly.jl`, verify on a Stage 1 smoke test before running stages 2/2m/3.

10.3 ForwardDiff AD Jacobian (Optional, Recommended for Hard Regions)

For regions where banded-FD truncation errors corrupt the Jacobian (cond(J) > 10^{15}, high AN depletion, etc.), use `ForwardDiff.jacobian!` instead of FD. Requires:

1. **Type-generic residual chain.** All functions in the residual call graph must accept `AbstractVector{<:Real}` or `::Real` scalars, not `::Float64`. Work arrays must use `zeros(eltype(u), ...)` not `zeros(...)`.
2. **Taylor-smoothed Bernoulli branch** in `sg_flux` (§3.3). A hard `if abs(α) < 1e-10` branch gives a spurious zero derivative at the α=0 seed; the Taylor form `B(α) = 1 − α/2 + α²/12 − ...` for |α| < 0.01 is smooth everywhere.
3. **ForwardDiff.jl + ForwardDiff.JacobianConfig** with a chunk size (default 12).

```
ad_cfg = ForwardDiff.JacobianConfig(residual!, zeros(n), u,
 ↪  ForwardDiff.Chunk{12}())
J = zeros(n, n)
ForwardDiff.jacobian!(J, residual!, zeros(n), u, ad_cfg)
```

AD cost is ~n/chunk_size forward passes at ~2× Float64 cost each — about 4× slower per Jacobian than banded FD, but **exact to machine precision** and identical to FD in easy regions (verify with a comparison test at startup).

10.4 Newton Solver — Direct `(J + λI) du = −F` with Strict L2 Descent

Direct Newton with small Tikhonov damping $\lambda = 10^{-10}$ (keeps near-singular Jacobians solvable without distorting the Newton direction). Strict L2 descent $||F\_new||_2 < ||F\_old||_2$ with 10-step backtracking line search. Step clamps per DOF type.

```
function newton_solve!(u, residual!;
                       tol = 1.0e-5, max_iter = 40,
                       lambda_fix = 1.0e-10,
                       jacobian_mode = :fd,              # :fd or :ad
                       max_backtrack = 10, verbose = false)
    # ... builds Jacobian (FD or AD), adds λ_fix to diagonal
    # ... solves (J + λI) du = -F
    # ... clamps du: log-conc <= 5.0, φ_l <= 0.015 V
    # ... backtracks α starting from 1 until merit decreases strictly
```

```
end
```

DOF type	Max step size per iteration
Log-concentration u_i	5.0 (natural log units)
Electrolyte potential ϕ_l	0.015 V

Convergence tolerance: **||F||_∞ < 10^{-5}**. This is loosened from 10^{-8} — charge conservation `|Σ z_i N_i|_interior < 10⁻¹⁰` is the real correctness gate (physicality check §13), and tight-tol wastes iterations chasing unreachable precision in near-singular regions.

> **Why not classical LM `(JᵀJ + λI)du = −JᵀF`?** Tested; heavy initial damping (`λ₀ = 1.0`) biased the continuation toward a non-physical solution branch. Direct `(J + λI)du = −F` with `λ = 10⁻¹⁰` preserves the Newton direction and stays on the physical attractor.

10.5 Continuation Strategy — V + Optional log-j Fallback

Simple Newton V-continuation (default). For each (δ, ε_org) pair, sweep V from −1.0 to −2.5 V vs SHE in steps of ds (initial 0.05 V), warm-starting Newton from the previous converged solution. Adaptive:

- Converges in <= 4 Newton iter -> grow ds by 1.4× (up to ds_max = 0.20 V)
- Converges in 5–10 iter -> grow by 1.1×
- Converges in 11–20 iter -> keep ds
- Converges in > 20 iter -> shrink by 0.7× (floor ds_min = 10^{-4} V)
- Fails (divergence, ||F|| not descending) -> shrink by 0.3×; retry

Optional log-j fallback via `newton_continuation_logj`. When V-continuation stalls at a near-fold, switch to parameterisation by $\log_{10}$(j_total) via augmented Newton with V as an extra DOF:

$$\text{augmented residual} = \begin{bmatrix} F(u,V) \\ \log_{10}(j_{\text{tot}}(u,V)) - \log_{10} j_{\text{target}} \end{bmatrix} = 0$$

Steps uniformly in log j. Equivalent to pseudo-arclength continuation with j as the arclength parameter. Primarily useful for dense sampling in the exponential-Tafel regime, not for extending the V-range (physical limits set the V floor).

10.6 Continuation Parameters α_buf and α_kin

Two scalar continuation parameters scale the source/flux terms during bootstrap (§12). Both are 0 at the start (residual is identically zero given the equilibrium initial guess) and ramped to 1 to recover the full physical model:

Parameter	Scales	Effect at α = 0	Effect at α = 1
α_buf	All buffer source terms R_buf,i	Buffer chemistry off — equilibrium IC is exact	Full buffer kinetics on
α_kin	All Faradaic flux densities j_1, j_2, j_3	No current; flat profiles preserved	Full Tafel kinetics on

10.7 Residual Form

> **Always use the integrated FV form** `F[ix] = J_left − J_right + S × dx` (residual in mol m^{-2} s^{-1}), not the divided form `(J_right − J_left)/dx + S`. On the graded mesh (dx ratio up to 10:1 at s = 10), the divided form makes residual entries O(1/dx_min), degrading Jacobian row scaling.

10.8 Full Residual Assembly (Type-Generic)

DOF layout: cell-major with 9 DOFs per cell. Helper inlines:

```
@inline conc_dof(ix, k) = 9*(ix-1) + k    # k = 1..8 (log-concentration DOFs)
@inline phi_dof(ix)     = 9*ix             # one per cell
```

The residual signature is **type-generic** for ForwardDiff compatibility:

```
function full_residual!(res::AbstractVector{T}, u::AbstractVector{T},
                        mesh, eps_org::Float64, V::Real,
                        alpha_buf::Real, alpha_kin::Real,
                        c_eq) where {T<:Real}
    N = length(mesh.dx)
    c   = zeros(T, 8, N)
    phi = zeros(T, N)
    for ix in 1:N, k in 1:8
        c[k, ix] = exp(clamp(u[conc_dof(ix, k)], -50.0, 50.0))
    end
    for ix in 1:N
        phi[ix] = u[phi_dof(ix)]
    end

    # SG fluxes at interior faces (2..N)
    J = zeros(T, 8, N+1)
    for ix in 1:N-1
        dx_face = 0.5 * (mesh.dx[ix] + mesh.dx[ix+1])
        for k in 1:8
            J[k, ix+1] = sg_flux(c[k, ix], c[k, ix+1],
                                 phi[ix], phi[ix+1],
                                 D_mix(k, eps_org), z_species[k], dx_face)
        end
    end
```

```
    # Faradaic BC at face 1 (electrode, x = 0)
    j1, j2, j3 = tafel_currents(c[6, 1], phi[1], V, alpha_kin)
    J[2, 1] = +(j1 + j2 + j3) / F          # OH⁻
    J[6, 1] = −(2*j1 + j2) / (2*F)         # AN
    J[7, 1] = +(j1) / (2*F)                # ADPN
    J[8, 1] = +(j2) / (2*F)                # PN
    # J[1, 1], J[3:5, 1] = 0 (H⁺, phosphates: no Faradaic flux)

    R_buf = zeros(T, 8)
    for ix in 1:N
        if ix == N
            # Bulk Dirichlet (gauge: φ_l(δ) = 0)
            for k in 1:8
                res[conc_dof(ix, k)] = u[conc_dof(ix, k)] −
                                       log(bulk_concentration(k, c_eq, eps_org))
            end
            res[phi_dof(ix)] = phi[ix]
        else
            buffer_sources!(R_buf, c[1,ix], c[2,ix], c[3,ix], c[4,ix], c[5,ix],
↪   alpha_buf)
            dx_ix = mesh.dx[ix]
            for k in 1:8
                res[conc_dof(ix, k)] = J[k, ix] − J[k, ix+1] + R_buf[k] * dx_ix
            end
           # Current conservation: Σ z_k (J_left − J_right) over k = 1..5 (charged)
            s_L = 0.0; s_R = 0.0
            for k in 1:5
                s_L += z_species[k] * J[k, ix]
                s_R += z_species[k] * J[k, ix+1]
            end
            res[phi_dof(ix)] = s_L − s_R
        end
    end
    return res
end
```

Note the bulk AN concentration comes from `bulk_concentration(6, c_eq, eps_org)` which calls `c_AN_bulk(eps_org)` per §7.2.

---

11. Solution Caching

Binary files encoding ε_org, δ, and V in the filename: `s_eo0.150_d50_V-1.300000.bin`. Format: Int64 DOF count + Float64 vector. Cache immediately after Newton converges.

---

12. Implementation Stages (All Mandatory)

Stage 1: NP + migration + buffer + kinetics at ε_org = 0.02 (single-phase reference).

Minimal physical AN loading. D_i,mix = D_i,aq for all species (single-phase regime). Validates migration, SG scheme, buffer chemistry, and electrode kinetics. **Not ε_org = 0 —** that's the pathological no-AN limit.

Sub-steps: 1. Compute `c_eq = solve_phosphate_equilibrium()` and build `u0 = make_initial_guess(N_mesh, c_eq, 0.02)`. At α_buf = 0, α_kin = 0, residual is identically zero. 2. **Buffer ramp:** α_buf from 0 -> 1 in 10 uniform steps at V = –1.0 V vs SHE, α_kin = 0. 3. **Kinetics ramp:** α_kin geometric ×2 from 10^{-6} to 1.0 (21 steps). Still at V = –1.0 V. 4. **Newton continuation sweep over V** from –1.0 to –2.5 V. 5. Cache -> plots -> **STOP for review.**

Stage 2: Activate D_i,mix(ε_org).

Same equations, but D_i,mix now depends on ε_org across both regimes. Run at multiple ε_org values {0.02, 0.05, 0.08, 0.15, 0.25, 0.30}. The 0.08 -> 0.15 transition directly probes the single-phase -> two-phase transition. Compare polarisation curves and FE_ADPN to Stage 1 to quantify the D_mix effect.

Sub-steps: for each ε_org, warm-start from the Stage 1 converged solution at V = –1.0 V -> Newton continuation sweep -> cache -> plots -> **STOP.**

Stage 3: Targeted forward sweep over the Core operating envelope (REVISED v6).

Stage 3 is no longer a "full 3D sweep over the v5 parameter cube". Its job is now to (a) **build a warm-start cache** and (b) **diagnose default-kinetics fit feasibility** before any optimisation runs. This stage produces no fitted parameters; it only produces predictions for visual inspection against Bloomquist.

Sub-steps: 1. **Warm-start cache.** Solve the model on a regular grid covering the Core envelope: `δ in {100, 130, 190, 220, 310 μm}` (covering Lévêque outputs for gap in {0.5, 1.0} mm × Q_total in {2, 6, 10}), `ε_org in {0.05, 0.08, 0.15, 0.20, 0.25, 0.30}`, and `V in [−1.0, −2.5] V vs SHE`. Cache converged solutions to `output/cache/`. 2. **Forward predictions on Core rows.** For each Core row (filter defined in §20.1), run `fixed_j_solver.jl` with default kinetics (j_0,r and α_c,r at §9.2 initial values) — read out (FE_ADN_model, FE_PN_model, V_cathode_SHE). 3. **Save** to `output/data/stage3_core_predictions.csv` aligned 1:1 with Core rows. 4. Generate fit-validation panels (§21 i, k, l) with default kinetics overlaid against Bloomquist Core. Inspect the *shape* of the curves: does FE_ADN rise with ε_org? drop with j past ε_sat? If shape is right, magnitudes are wrong -> kinetics fit is meaningful (proceed to Stage 4). If shape is wrong -> revisit kinetics form or m_i diffusivity (§4.2) before fitting. 5. **STOP for review.**

Stage 4: Bloomquist kinetics-only fit (REVISED v6).

Kinetics-only fit against the *Core* Bloomquist subset with transport frozen and `(V_CE, R_contact)` frozen at literature defaults. Sub-steps:

1. **Stage 4a — Core fit.** Optimise `(j₀,1, j₀,2, j₀,3, α_c,1, α_c,2, α_c,3)` against Core rows (~60 rows; filter §20.1). Use Levenberg–Marquardt warm-started from §9.2 defaults. Loss = Σ (FE_model − FE_exp)² on `(FE_ADN, FE_PN)`. V_CE = 1.7 V, R_contact = 1×10^{-4} Ω·m² **frozen** — not fit (see §20.2).
2. **Stage 4b — forward apply (no re-fit).** Apply Stage 4a's converged params to (i) Extended subset (~96 rows: gap in {0.5, 1.0} mm, full j and ε_org range), then (ii) Full holdout (54 rows:

gap = 0.25 mm). Save residuals to `output/data/stage4b_extended_residuals.csv` and …`holdout_residuals.csv`.

3. **Decision gates** (Stage 4b results):
   - Core RMSE > 12 pp on FE_ADN -> kinetics-form bug; do not proceed.
   - Core RMSE < 8 pp **and** Extended RMSE > 12 pp -> bubble physics matters; v7 work scoped.
   - Holdout RMSE > 15 pp on FE_ADN -> bubble physics is required for 0.25 mm gap.
4. Generate fit-validation panels (§21 i–o) with three series overlaid (Core / Extended / Holdout). **STOP for review.**

Stage 4c (optional, deferred to v7). A joint refinement on `(j₀,r, α_c,r, V_CE, R_contact)` over Core + Extended is *not* part of v6. Reason: v6 has no honest `V_cell` residual to fit `V_CE` and `R_contact` against — Bloomquist tabulates `EP_ADN` (energy productivity) but not per-row V_cell, so any V_CE / R_contact fit in v6 would be against a back-derived noisy quantity. Defer to v7 alongside bubble physics.

---

13. Physicality Checks

Check	Expected
ϕ_l(x) profile	A few mV variation; larger at high j
Σ z_i J_i at each face	= 0 to machine precision
9-species electroneutrality |Σ z_i c_i|	< 10^{-8} mol m^{-3} (trivially 0 by construction)
c_AN(x) at high j	Depletes toward electrode
Buffer residuals at bulk	< 10^{-6} mol m^{-3} s^{-1} (Dirichlet exact)
pH(x)	Rises monotonically from bulk to surface at high j (OH^--pathway)
FE_ADPN vs ε_org	Peaks around ε_sat ~= 0.086 up in two-phase regime
D_AN,mix vs ε_org	Flat (single-phase) then rises (two-phase)
D_OH,mix vs ε_org	Flat (single-phase) then drops (two-phase)
No R_PT residual	Confirm R_PT is absent from residual
Bulk pH at x = δ	~= 13.03 (matches solve_phosphate_equilibrium with updated Ka)

---

14. Module Structure

v7 model code lives in `an_ehd_v2/`, sibling of the v6 baseline at `an_ehd/`. The experimental data was moved from `an_ehd/Experimental_data/` to a project-root-level `Experimental_data/` so both versions share the same source CSVs.

```
ADPN-Julia-Model/
```

```
├── an_ehd/                                # v6 / v6.x baseline — frozen
│   ├── (same module structure as v7 below)
│   └── output/stage4/, stage4v2/          # frozen v6 + v6.x fit baselines
│
├── an_ehd_v2/                             # v7 model code (NEW)
│    ├── ADPN_EHD.jl                         # Master module — includes & re-
exports all submodules
│   ├── params.jl                  # Constants extended to 9-species v7 layout
│    ├── mesh.jl                           # make_mesh(N, delta; stretch)
│   ├── diffusivity.jl                  # D_mix length-9 arrays (TCH appended)
│  ├── chemistry.jl            # solve_phosphate_equilibrium, buffer_sources!,
│  │                        # c_AN_bulk, make_initial_guess (extended for TCH)
│     ├── kinetics.jl                                 # tafel_currents -> 4-
tuple; KIN_OVERRIDE carries
│   │                           # j0::NTuple{4}, ac::NTuple{4}, n::NTuple{3}
│   ├── transport.jl                  # sg_flux with Taylor-smoothed Bernoulli
│   ├── assembly.jl                   # full_residual! with 10-DOF/cell layout
│    ├── solver.jl                        # newton_solve!; newton_continuation;
│  │                      # newton_continuation_logj; galv_continuation (NEW)
│     ├── sweep_runner.jl                         # mesh -> bootstrap -> V-
cont pipeline (Stages 1–3)
│    ├── hydrodynamics.jl                   # delta_leveque, weber_numbers (§18)
│  ├── cell_voltage.jl         # kappa_eff, R_series, V_cell_predicted (§17)
│     ├── fixed_j_solver.jl                       # solve_at_j with n_orders 3-
tuple kwarg (v7)
│   ├── fit_kinetics.jl              # N_THETA = 9; HER frozen; FE_TCH residual
│      ├── run_stage1/2/2m/3.jl                         # cache rebuild (v7 9-
species DOF breaks v6 cache)
│  ├── run_stage4v3.jl         # NEW v7 — TCH-aware fit, output to stage4v3/
│   ├── analyze_stage4.jl                # extended to read 9-key fitted-theta
│   ├── plot_stage4v3_*.py              # parity / residual / regime-map plots
│   └── output/
│         ├── cache/                       # v7 9-species warm-start states
│         └── stage4v3/{data,logs,plots}/  # v7 fit results
│
├── Experimental_data/            # NEW: shared between an_ehd/ and an_ehd_v2/
│    ├── bloomquist_data.csv             (162 rows × 14 cols, master)
│    └── Table_S2..S10_*.csv             (per-block source tables)
│
└── Guide Docs/                          # implementation guides + changelogs
```

14.1 KIN_OVERRIDE Ref (extended v7)

The `KIN_OVERRIDE` Ref in `kinetics.jl` is the central plumbing for variable kinetics. v7 extends its tuple shapes:

```
# v6 KIN_OVERRIDE
@NamedTuple{j0::NTuple{3,Float64},    # (ADPN, PN, HER)
            ac::NTuple{3,Float64}}    # (ADPN, PN, HER)

# v6.x KIN_OVERRIDE — adds n
@NamedTuple{j0::NTuple{3,Float64},
            ac::NTuple{3,Float64},
            n ::NTuple{2,Float64}}    # (n_ADN, n_PN)

# v7 KIN_OVERRIDE — adds TCH branch
@NamedTuple{j0::NTuple{4,Float64},    # (ADPN, PN, HER, TCH)
            ac::NTuple{4,Float64},    # (ADPN, PN, HER, TCH)
            n ::NTuple{3,Float64}}    # (n_ADN, n_PN, n_TCH)
```

`tafel_currents` returns a 4-tuple `(j_1, j_2, j_3, j_4)`. The HER branch ignores n. Default fallback (override = `nothing`) reads from Params with hard-coded `n_ADN = 2, n_PN = 1, n_TCH = 3` so Stages 1/2/2m/3 stay byte-identical to v5/v6 in their default-kinetics paths.

14.2 Notes

- `solve_phosphate_equilibrium` uses an inline bisection (not `Roots.jl`) to avoid Windows Defender Application Control blocking `Roots`'s DLL cache on some systems.
- The 9-species DOF layout invalidates every v6 cache. Stages 1, 2, 2m, 3 must be re-run before Stage 4v3 in `an_ehd_v2/`.
- v6 `an_ehd/run_stage4.jl` and `run_stage4v2.jl` are gated with `error()` to prevent overwriting the frozen v6 / v6.x baselines.
- `run_stage4v3.jl` writes to `an_ehd_v2/output/stage4v3/`.
- Experimental-data path inside `an_ehd_v2/` files becomes `joinpath(@__DIR__, "..", "Experimental_data", "bloomquist_data.csv")` (one directory up).

---

15. Common Pitfalls

Pitfall	Symptom	Fix
ε_org = 0 in sweep	c_AN,bulk = 0, log(0) = –Inf, NaN residual	Sweep from ε_org >= 0.02
Using D_aq everywhere	ε_org has no effect past ε_sat	Use regime-aware D_mix
D_org ≠ 0 for ions	Unphysical ionic transport	Set D_org = 0 for all ions
Single-phase with mixed D	Spurious organic pathway when no droplets exist	D_mix = D_aq for ε_org < ε_sat
Adding ε_aq to R_buf	Double-counting	No prefactor (c_i is aqueous)

Pitfall	Symptom	Fix
Including R_PT	Inconsistent with local equilibrium	Remove R_PT entirely
Wrong AN order (v6 default)	FE insensitive to j	v7 promotes n_ADN, n_PN, n_TCH to fit params; default fallback retains 2/1/3
H^+-pathway buffer at high pH	Unphysical k_r ~ 10^{10} $m^3/(mol \cdot s)$	OH^--pathway stoichiometry (§6)
Cold start (no equilibrium IC)	Newton diverges immediately	solve_phosphate_equilibrium + make_initial_guess
SG overflow	exp crash	Clamp α to [–700, 700]
SG hard branch at α=0	AD derivative spuriously 0	Taylor expansion for $\lvert\alpha\rvert < 0.01$
Arithmetic D too weak	FE enhancement too small	Upgrade to m_i-corrected D_eff (§4.2)
Divided FV form on graded mesh	Jacobian ill-conditioned	Use integrated form J_L – J_R + S·dx (§10.7)
Missing ϕ_l gauge fix	Jacobian rank-deficient, Newton stalls	Set ϕ_l(δ) = 0 (Dirichlet, §7.2)
j_0 in mA/cm^2 passed directly	Rates off by factor of 10	Convert: $A/m^2 = mA/cm^2 \times 10$
Dense FD Jacobian	~60 ms/step, full sweep >10 min	Banded FD with 35-color grouping + sparse LU (§10.2)
Jacobian halfbandwidth b = 9	Banded FD aliases columns, exactly singular	Use b = 2·block_size – 1 = 17 (§10.2)
Fixed absolute dx_min/dx_max in mesh	Domain length wrong across δ sweep	Use stretch-factor parameterisation
ϕ_l step clamp at 0.1 V	Over-large potential steps destabilise c_i DOFs	Clamp ϕ_l steps to 0.015 V
Implementing PAC before testing Newton cont.	Unnecessary complexity	Use simple Newton continuation first (§10.5)

Pitfall	Symptom	Fix
Classical LM `(JᵀJ+λI)du=−JᵀF` with λ_0=1	Picks wrong solution branch	Use direct `(J+λI)du=−F` with $\lambda=10^{-10}$ (§10.4)
Merit slack 1.10 at tight tol	Accepts numerical-noise as progress	Strict L2 descent at tol <= 10^{-5}
Non-type-generic residual	ForwardDiff fails with type error	`AbstractVector{T<:Real}`; `zeros(eltype(u),...)`
(v6) j_0 in mA/cm² fed to Tafel	Currents off ×10	Convert in `params.jl`: A/m² = mA/cm² × 10 (already in v5; re-flagged for fit code)
(v6) Fitting against FE_TCH	TCH not in model — silent over-fit	**Resolved in v7**: TCH is now species 9 with its own Tafel reaction; FE_TCH is a fit residual channel
(v7) Hardcoded 8 in array indexing	Off-by-one when extending to 9 species	Search codebase for literal 8; use `Params.n_species`
(v7) AD type stability through 4-tuple `tafel_currents`	ForwardDiff specialises on tuple length	Rebuild types from a fresh Julia session after updating `assembly.jl` callers
(v7) Bandwidth not updated 17 -> 19	Banded-FD aliases columns, exactly singular	Update `JAC_BLOCK = 10`, `JAC_HALFBW = 19`, `n_colors = 39` everywhere
(v7) TCH MW mismatch (161 vs 175)	FE_TCH residuals biased ×0.92 if Bloomquist used saturated MW	Confirm Bloomquist's FE_TCH column uses $C_9H_{11}N_3$ (n_e=2) MW = 161.20

Pitfall	Symptom	Fix
(v7) `Experimental_data` path stale	"file not found" when running from `an_ehd_v2/`	Use `joinpath(@__DIR__, "..", "Experimental_data", ...)` (parent dir)
(v7) HER frozen value mismatch with `Params`	Inconsistent results between fit driver and Stages 1-3	Update Params HER constants to v6.x values OR apply permanent override; pick one
(v6) Fitting on 0.25 mm gap rows	Bubble physics absent -> biased kinetic params	Use 0.5 + 1.0 mm rows for training; 0.25 mm is holdout (§20.3)
(v6) V_cell sign confusion	`V_cathode_SHE` is negative; `V_cell` is a positive magnitude	Use `abs(V_cathode_SHE)` in V_cell_predicted; verify with §17.1 sign convention block
(v6) Forgetting Bruggeman correction on κ	κ overpredicted at high ε_org -> R_series too small	`κ_eff = κ_dilute · (1 - ε_org)^1.5` (§17.2)
(v6) Lévêque applied above transition Re	Sh correlation invalid past Re ~= 2000	All Bloomquist rows have Re < 10 — Lévêque is safe; flag if extending the dataset
(v6) Newton continuation stuck on fixed-j root	Bisection on V wraps around fold	Bracket V in [–2.5, –0.8] V; bisect with monotone j(V) verified at startup
(v6) δ_lam computed using Q_aq instead of Q_total	Underestimates BL by ~20%	Lévêque uses superficial velocity from total volumetric throughput, both phases

16. Potential Referencing

Model works internally in V vs SHE. Onsets from Mathison et al. (JACS 2025) on Cd: AN reduction −1.28 V; optimal ADPN −1.62 V; HER (no AN) −1.43 V.

Reference electrode	Conversion
Ag/AgCl (sat. KCl)	E_SHE = E + 0.197 V
SCE	E_SHE = E + 0.241 V
RHE	E_SHE = E − 0.059 × pH (25°C)

---

17. Cell-Voltage Decomposition (NEW v6)

The model solves for `V_cathode` vs SHE; Bloomquist reports two-electrode `V_cell` (a positive magnitude). To compare, the unobserved components — anode, bulk-electrolyte ohmic drop, contact resistance — must be added back in. v6 uses a single lumped fit term `V_CE` for the anode and computes the bulk-electrolyte term from the model's own bulk composition. **No bubble void correction in v6** (deferred to v7); the 0.25 mm gap data is therefore expected to systematically underpredict V_cell.

17.1 Decomposition

```
V_cell = V_CE − V_cathode_SHE + j · R_series
R_series = (gap − δ) / κ_eff(c_bulk, ε_org) + R_contact
```

Equivalently (the form used in the fitting loop):

$$V_{\text{cathode SHE}} = V_{CE} - V_{\text{cell,meas}} + j\left[(\text{gap} - \delta)/\kappa_{\text{eff}} + R_{\text{contact}}\right]$$

Term	Source	Status in v6
V_CE	Lumped anode contribution: E°_OER + on SS	**Fit scalar**, range 1.5–2.0 V (initial guess 1.7 V)
V_cell,meas	Bloomquist `V_cell` (positive magnitude — paper convention)	Not directly tabulated in S2–S10; not used in v6 fits because the SI omits per-row V_cell. Energy productivity gives `V_cell` indirectly: `V_cell = (j · M_ADN) / (n_e · F · PR_ADN / EP_ADN)` if needed.

Term	Source	Status in v6
(gap – δ)/κ_eff	Bulk electrolyte ohmic drop in the gas-bubble-free portion of the gap	**Computed** from §17.2
R_contact	Spring-probe + foil contact resistance	**Fit scalar**, range 0.5×10^{-4} – 2×10^{-4} Ω·m² (initial 1×10^{-4})
Cathode kinetic + concentration overpotentials	Already in `V_cathode_SHE` via Tafel + diffusion-layer ϕ_l(0)	**Do not double-count**
Membrane drop	None (undivided cell)	Skip
Anode-side mass-transport + bubble	Lumped into V_CE for v6	Deferred
Cathode H_2 + anode O_2 bubble void	None	**Deferred to v7**

Sign convention. Bloomquist's `V_cell` is the magnitude of the applied two-electrode bias (positive). The cathode is at the negative side, so `V_cathode_SHE` is negative (~ −1 to −2.5 V). With V_CE positive (~1.7 V), the equation reduces to a magnitude balance: `|V_cell| = V_CE + |V_cathode_SHE| + j·R_series`.

17.2 Bulk Electrolyte Conductivity from Composition

In v6 we compute κ_eff directly from the model's bulk-equilibrium composition — **no fit parameter for solution conductivity.** Dilute-solution theory (Newman, Electrochemical Systems §11.3):

$$\kappa_{\text{dilute}} = \frac{F^2}{RT}\sum_i z_i^2\, D_{i,\text{aq}}\, c_{i,\text{bulk}}$$

evaluated at x = δ where the model is well-mixed and `c_i,bulk` comes from `solve_phosphate_equilibrium()` (§6.4). Sum runs over all charged species: H^+, OH^-, $H_2PO_4^-$, HPO_4^{2-}, PO_4^{3-}, Na^+. The Bruggeman porosity correction accounts for the organic-droplet volume blocking ionic conduction:

$$\kappa_{\text{eff}}(\varepsilon_{\text{org}}) = \kappa_{\text{dilute}}\cdot(1-\varepsilon_{\text{org}})^{1.5}$$

```
function kappa_dilute(c_eq)
    # c_eq fields: H, OH, H2PO4, HPO4, PO4, Na  [mol/m³]
    # D_aq for ions: H⁺ 9.31e-9, OH⁻ 5.27e-9, H₂PO₄⁻ 0.846e-9,
    #                HPO₄²⁻ 0.690e-9, PO₄³⁻ 0.610e-9, Na⁺ 1.33e-9
    coeff = F^2 / (R_gas * T)
    return coeff * (
        1^2 * 9.31e-9 * c_eq.H    +
        1^2 * 5.27e-9 * c_eq.OH   +
        1^2 * 0.846e-9 * c_eq.H2PO4 +
        2^2 * 0.690e-9 * c_eq.HPO4  +
```

```
        3^2 * 0.610e-9 * c_eq.PO4   +
        1^2 * 1.33e-9 * c_eq.Na
    )
end
kappa_eff(c_eq, eps_org) = kappa_dilute(c_eq) * (1.0 - eps_org)^1.5
```

Sanity check at the v6 base composition (0.5 M Na_3PO_4 + 0.02 M TBA-OH, pH = 13.03, c_Na = 1520 mol/m³). Numbers below are from the actual `cell_voltage.jl` evaluation (not a sketch):

Quantity	Value	Unit
κ_dilute	19.1	S m^{-1}
κ_eff (ε_org = 0.02)	18.6	S m^{-1}
κ_eff (ε_org = 0.0862, threshold)	16.7	S m^{-1}
κ_eff (ε_org = 0.15)	15.0	S m^{-1}
κ_eff (ε_org = 0.30)	11.2	S m^{-1}

This is **higher** than the empirical 5–10 S/m range proposed in CONTEXT_TRANSFER §7, primarily because PO_4^{3-} contributes $z^2 = 9$ weighting at pH = 13 (most phosphate sits as PO_4^{3-}, not HPO_4^{2-}). Concentrated-solution ion pairing is expected to reduce this by ~20–40% in reality, but v6 deliberately stays with dilute theory; if Stage 4b residuals show systematic bias correlated with j, this is the first place to revisit (treat κ as a tier-2 fit param in v7).

> **Why the diffusion-layer ohmic drop is *not* in (gap – δ)/κ_eff.** The Nernst–Planck solver already integrates `dφ_l/dx` from `x = 0` to `x = δ` as part of the current-conservation equation (§3.2). That contribution is inside `V_cathode_SHE` via the ϕ_l(0) value at the electrode face. The (gap – δ) term covers only the *unmodeled* bulk between the diffusion-layer edge and the anode.

17.3 Module: `cell_voltage.jl`

```
module CellVoltage
using ..Params, ..Chemistry

function kappa_dilute(c_eq)
    coeff = F^2 / (R_gas * T)
    z2D = (
        1.0 * 9.31e-9 * c_eq.H,
        1.0 * 5.27e-9 * c_eq.OH,
        1.0 * 0.846e-9 * c_eq.H2PO4,
        4.0 * 0.690e-9 * c_eq.HPO4,
        9.0 * 0.610e-9 * c_eq.PO4,
        1.0 * 1.33e-9 * c_eq.Na,
    )
    return coeff * sum(z2D)
end

kappa_eff(c_eq, eps_org) = kappa_dilute(c_eq) * (1.0 - eps_org)^1.5
```

```
# Map model output to predicted cell voltage magnitude
function V_cell_predicted(V_cathode_SHE, j, gap, delta, eps_org, c_eq;
                          V_CE, R_contact)
    R_series = (gap - delta) / kappa_eff(c_eq, eps_org) + R_contact
    return V_CE + abs(V_cathode_SHE) + j * R_series   # positive magnitude
end

# Inverse: given measured V_cell, return the V_cathode the model should solve at
function V_cathode_target(V_cell_meas, j, gap, delta, eps_org, c_eq;
                          V_CE, R_contact)
    R_series = (gap - delta) / kappa_eff(c_eq, eps_org) + R_contact
    return -(V_cell_meas - V_CE - j * R_series)       # signed (negative)
end
end
```

Inputs `j` in A m^{-2}, `gap`, `delta` in m, `R_contact` in Ω m^2. Output V in V.

---

18. Hydrodynamics: Flow -> δ Mapping (NEW v6)

Bloomquist parameterises by interelectrode gap × total flow rate × phase split; the model needs δ. v6 uses the laminar Lévêque correlation with no bubble-induced enhancement (deferred to v7). The We numbers are reported alongside δ for diagnostic flow-regime classification but do not feed back into the model.

18.1 Channel Geometry (Bloomquist reactor)

Symbol	Value	Source
Active electrode area A	6.4 cm^2	SI §"Reactor Setup"
Channel width W	4 mm = 4×10^{-3} m	SI §"Reactor Setup" (estimated; serpentine)
Channel length L	A / W = 16 cm = 0.16 m	Derived
Gap (interelectrode)	0.25, 0.5, 1.0 mm	Sweep parameter
Hydraulic diameter d_h	2·gap·W/(gap+W)	Rectangular duct

> Channel width W comes from the FEP gasket cut. If the fitted V_cell residuals show a systematic gap-dependent bias, treat W as a tier-2 fit parameter; otherwise leave fixed at 4 mm.

18.2 Lévêque Boundary Layer

Superficial velocity, kinematic viscosity, Reynolds, Schmidt:

```
v = Q_total / (gap · W)            [m/s], Q_total in m³/s
ν = 1×10⁻⁶ m²/s                     (water-like, 25 °C)
```

```
Re = v · d_h / ν
Sc = ν / D_AN,aq ~= 1×10⁻⁶ / 2.30×10⁻⁹ ~= 435
```

Lévêque correlation for laminar developing concentration boundary layer in a duct:

$$\text{Sh} = 1.85\left(\text{Re}\cdot\text{Sc}\cdot d_h/L\right)^{1/3}$$

$$\delta_{\text{lam}} = d_h/\text{Sh}$$

In v6 we set δ = δ_lam directly. (CONTEXT_TRANSFER §7 proposed a K_δ geometric correction in [0.3, 3.0]; v6 fixes K_δ = 1.0. If FE residuals show a systematic Q-dependent bias after kinetic fitting, promote K_δ to a tier-2 fit parameter in v7 alongside bubble effects.)

Sanity-check table (W = 4 mm, L = 16 cm, ν = 10^{-6} m²/s, D_AN = 2.3×10^{-9} m²/s, Sc ~= 435). Values are from the actual `hydrodynamics.jl` evaluation:

gap [mm]	Q_total [mL/min]	v [cm/s]	δ_lam [μm]
0.25	2	3.33	93.6
0.25	6	10.0	64.9
0.25	10	16.67	54.7
0.5	2	1.67	145.8
0.5	6	5.00	101.1
0.5	10	8.33	85.3
1.0	2	0.83	223.5
1.0	6	2.50	154.9
1.0	10	4.17	130.7

δ_lam ranges 55–224 μm across the 9 (gap × Q_total) blocks. v6's pre-existing δ sweep grid {10, 20, 50, 100, 200 μm} no longer covers the high-gap / low-flow corner (224 μm) — Stage 3 cache should be re-built on a δ grid driven by the Lévêque outputs above (see §12 Stage 3 sub-step 1).

18.3 Weber Numbers (diagnostic only in v6)

For each phase:

$$\text{We}_i = \frac{\rho_i\, v_i^2\, \text{gap}}{\sigma_{\text{AN—water}}}$$

with phase superficial velocities `v_i = Q_i / (gap · W)`, ρ_aq = 1000 kg/m³, ρ_org = 810 kg/m³, σ_AN-water = 10.5 mN/m (Girifalco–Good, SI §"Weber Number Calculations"). These are reported in the Bloomquist tables and serve as flow-regime coordinates (droplet / slug / parallel

/ transition). In v6 they are computed for diagnostic plotting and CSV export only — they do not enter the model.

> **Cross-check.** Recomputing We_aq, We_org from the Q_aq, Q_org, gap columns in `bloomquist_data.csv` should reproduce the tabulated values to within rounding (the SI rounds to 1–2 sig figs). Add this as a unit test in `hydrodynamics.jl`.

18.4 Module: `hydrodynamics.jl`

```julia
module Hydro
const W_CHANNEL = 4.0e-3       # m, FEP gasket width
const L_CHANNEL = 0.16         # m, serpentine path length (= A/W)
const NU_KIN    = 1.0e-6       # m²/s, water at 25°C
const RHO_AQ    = 1000.0       # kg/m³
const RHO_ORG   = 810.0        # kg/m³
const SIGMA_AN  = 10.5e-3      # N/m, AN-water interfacial tension

d_hydraulic(gap) = 2 * gap * W_CHANNEL / (gap + W_CHANNEL)
v_super(Q_m3s, gap) = Q_m3s / (gap * W_CHANNEL)

function delta_leveque(gap_m, Q_total_m3s; D_ref = 2.30e-9)
    d_h = d_hydraulic(gap_m)
    v   = v_super(Q_total_m3s, gap_m)
    Re  = v * d_h / NU_KIN
    Sc  = NU_KIN / D_ref
    Sh  = 1.85 * (Re * Sc * d_h / L_CHANNEL)^(1/3)
    return d_h / Sh
end

function weber_numbers(gap_m, Q_aq_m3s, Q_org_m3s)
    v_aq  = v_super(Q_aq_m3s,  gap_m)
    v_org = v_super(Q_org_m3s, gap_m)
    We_aq  = RHO_AQ  * v_aq^2  * gap_m / SIGMA_AN
    We_org = RHO_ORG * v_org^2 * gap_m / SIGMA_AN
    return (We_aq=We_aq, We_org=We_org)
end

# Q convenience: mL/min -> m³/s
ml_min_to_m3_s(q) = q * 1.0e-6 / 60.0
end
```

> **Bubble-correction stub for v7.** When v7 lands, replace `delta_leveque` with `delta_actual = delta_leveque · f_bubble(j, gap, Q)` where `f_bubble` is the bubble-induced convection enhancement (Vogt 1983 or fitted directly). This is the dominant missing physics for the 0.25 mm gap rows.

---

19. Experimental Data

19.1 Bloomquist et al. (CEJ 2026, 528, 172125) — primary fitting target

Setup: parallel-plate undivided flow reactor, Cd-foil cathode, SS anode, 0.5 M Na_3PO_4 + 0.02 M TBA-OH + 0.03 M EDTA, T = 25 °C. Active area 6.4 cm^2. 162 Hammersley-sampled experiments. Headline results: FE_ADPN = 73–76% maintained at j > 200 mA cm^{-2} when ε_org > ε_sat, with bubble-induced convection (not flow regime) the dominant transport enhancer.

Dataset: `an_ehd/Experimental_data/bloomquist_data.csv` — 162 rows × 14 columns, plus per-table CSVs `Table_S2`…`Table_S10.csv` for the original 9 (gap × Q_total) blocks.

Column	Symbol	Unit	Notes
`table`	—	—	One of S2–S10
`gap_mm`	gap	mm	0.25, 0.5, or 1.0
`Q_total_mL_min`	Q_total	mL/min	2, 6, or 10 (sum of phases)
`j_mA_cm2`	j	mA/cm^2	Applied current density (×10 -> A/m^2)
`phi_AN`	ε_org	—	AN volume fraction (= Q_org / Q_total)
`Q_aq_mL_min`	Q_aq	mL/min	Aqueous-phase flow
`Q_org_mL_min`	Q_org	mL/min	Organic-phase flow
`We_aq`	We_aq	—	Aqueous Weber, recomputable
`We_org`	We_org	—	Organic Weber, recomputable
`FE_ADN_pct`	FE_ADPN	%	**Primary fit target**
`FE_TCH_pct`	FE_TCH	%	Tricyanohexane (trimer); **fit target in v7** (§19.3)
`FE_PN_pct`	FE_PN	%	Propionitrile, fit target
`PR_ADN_kg_cm2_h`	PR_ADPN	kg cm^{-2} h^{-1}	Production rate, derived from FE × j
`EP_ADN_kg_kWh`	EP_ADPN	kg kWh^{-1}	Energy productivity — implicitly contains V_cell

V_cell back-out. The SI tables omit V_cell directly. It can be recovered per row from EP_ADN: `V_cell = (M_ADN · j · A · 3600) / (n_e · F · PR_ADN_per_area / EP_ADN)`. Use for diagnostic only — don't fit V_cell residuals against a back-derived quantity.

19.2 FE_HER recovery

Bloomquist tables report FE_ADN, FE_TCH, FE_PN. The remainder is HER + side products: `FE_HER ~= 100% − FE_ADN − FE_TCH − FE_PN` (assuming no losses to oligomers/heavies beyond TCH). v6 model output gives FE_HER directly; compare with this back-derived value.

19.3 TCH in v7 (was deferred in v6)

TCH (1,3,6-tricyanohexane) is the AN-trimer side product whose 5–17% experimental FE was previously absorbed into the model's FE_HER residual in v6. **v7 adds TCH as species 9 with its own Tafel reaction** (§5 R4), `n_e_TCH = 2` from the Bloomquist SI, three new fit parameters `(j₀,TCH, α_c,TCH, n_TCH)`, and an FE_TCH residual term in the loss function. The conserved quadruple `FE_ADN + FE_PN + FE_HER + FE_TCH = 100%` replaces the v6 triplet, giving cleaner FE_HER comparison.

19.4 Other references (unchanged from v5)

Mathison et al. (JACS 2025, 147, 4296): Mechanism — radical coupling for ADPN, proton transfer for PN (strong KIE). **Suwanvaipattana et al.** (J. Cleaner Prod. 2017, 142, 1296): D, m, d_p values, AN density (806 kg/m³). **Huang et al.** (CEJ 2020, 382, 123006): Langmuir-adsorption kinetics on Pb. **Costentin & Savéant** (J. Electroanal. Chem. 564, 2004, 99): $\Delta G^0 = -1.84$ eV for protonated radical coupling. **Eigen & De Maeyer** (Z. Elektrochem. 1955): water autoprotolysis rate k_1,f = 1.4 × 10^{-3} M/s = 1.4 mol/(m³·s). **Newman** (Electrochemical Systems, 3rd ed., §11.3): dilute-solution conductivity κ = (F²/RT)·Σ z_i²·D_i·c_i. **Lévêque** (Ann. Mines 1928); **Bird, Stewart & Lightfoot** §14.4: developing-BL Sh = 1.85·(Re·Sc·d_h/L)^(1/3).

---

20. Fitting Strategy (kinetics-only, transport frozen)

v7 fits **nine kinetic parameters** `(j₀,1, j₀,2, j₀,TCH, α_c,1, α_c,2, α_c,TCH, n_ADN, n_PN, n_TCH)` (was 6 in v6, 8 in v6.x). **All transport quantities are computed, not fit:** δ from Lévêque (§18), κ_eff from bulk composition (§17.2), D_i,mix from §4. The two cell-voltage scalars `(V_CE, R_contact)` and the HER kinetics `(j₀,3, α_c,3)` are **frozen** — see §20.5 and §9.2 for why.

20.1 Row selection — Core / Extended / Holdout

Bloomquist provides 162 rows. v6 partitions them into three concentric subsets ordered by trust in v6 physics:

Subset	Filter	Rows	Used for
Core	gap in {0.5, 1.0} mm AND j <= 190 mA cm^{-2} AND ε_org >= 0.04	~=60	Stage 4a fit

Subset	Filter	Rows	Used for
Extended	gap in {0.5, 1.0} mm AND ε_org >= 0.04 (any j)	~=96	Stage 4b forward apply (no re-fit)
Full holdout	gap = 0.25 mm AND ε_org >= 0.04	~=48	Stage 4b forward apply, untouched during fitting

Why these filters. The 0.25 mm gap holdout isolates rows where bubble void blocking dominates (v6 has no bubble physics, so these will systematically miss). The j <= 190 mA cm^{-2} Core cap excludes the bubble-convection-dominated high-j regime where v6 will systematically under-predict mass transport. The ε_org >= 0.04 filter drops rows where AN is starved (FE_ADN ~= 0 experimentally) — they are physically degenerate and add noise without information. Bloomquist has 18 rows with ε_org < 0.04 (~= 11% of the dataset), all dropped.

What "ε_org >= 0.04" actually filters. Each (gap, Q_total) block has 18 rows; 2 of those 18 are ε_org = 0.02. So the ε_org filter drops 18/162 = 11% of rows. Combined with the gap and j filters above: - Core: 6 (gap × Q_total) blocks × ~10 rows after j and ε_org filters ~= **60 rows** - Extended: 6 (gap × Q_total) blocks × 16 rows after ε_org filter = **96 rows** - Holdout: 3 (gap × Q_total) blocks × 16 rows after ε_org filter = **48 rows**

20.2 Workflow

For each retained row with `(gap, Q_total, ε_org, j, FE_ADN, FE_PN)`:

1. **Pre-compute transport.** Q_aq, Q_org -> δ_lam via `delta_leveque(gap, Q_total)`. ε_org -> κ_eff via §17.2. Cache per (gap, Q_total, ε_org) tuple.
2. **Solve at fixed j.** Bloomquist runs at constant current; the model solves at constant V. Use `fixed_j_solver.jl`: bisect V vs SHE until `Σ j_r(V) = j_target` to within 1 mA cm^{-2}, read out (FE_ADN, FE_PN, V_cathode_SHE). Warm-start V from the nearest cached `(gap, Q_total, ε_org, V)` solution.
3. **Predict V_cell** (diagnostic only, not in loss). `V_cell_pred = V_CE + |V_cathode_SHE| + j · R_series` (§17). Plot against `EP_ADN`-derived V_cell back-out (§19.1) for a sanity check.
4. **Loss** (Stage 4a only — Core rows N_core ~= 48 in v7; FE_TCH channel added):

$$\mathcal{L}(\theta) = \sum_{r \in \text{Core}} \sum_{p \in \{\text{ADN,PN,TCH}\}} \left(\text{FE}^{\text{model}}_{p,r}(\theta) - \text{FE}^{\text{exp}}_{p,r}\right)^2$$

5. **Optimiser.** Levenberg–Marquardt with finite-difference Jacobian on θ in R^9. Initial guess from §9.2. Bounds enforced via log-transform on j_0,r and box clipping on α_c,r and n_r. Expected: 10–20 LM iterations × ~1 s per fixed-j solve × ~48 Core rows × 3 species ~= **30–60 min wall time** (with Stage 3 warm-start cache).

20.3 Fit parameters (v7)

Parameter	Initial	Bounds	Unit	Status
j_0,1 (ADPN)	1×10^{-3}	[10^{-6}, 10^{-1}]	A m^{-2}	**fit**
j_0,2 (PN)	1×10^{-3}	[10^{-6}, 10^{-1}]	A m^{-2}	**fit**
j_0,3 (HER)	**2.666×10^{-5}**	—	A m^{-2}	**frozen at v6.x converged**
j_0,TCH (NEW v7)	**1×10^{-3}**	**[10^{-6}, 10^{-1}]**	**A m^{-2}**	**fit**
α_c,1	0.5	[0.3, 0.7]	—	**fit**
α_c,2	0.5	[0.3, 0.7]	—	**fit**
α_c,3	**0.390**	—	—	**frozen at v6.x converged**
α_c,TCH (NEW v7)	**0.5**	**[0.30, 0.70]**	—	**fit**
n_ADN (NEW v6.x)	2.0	[1.0, 3.0]	—	**fit**
n_PN (NEW v6.x)	1.0	[0.5, 2.0]	—	**fit**
n_TCH (NEW v7)	3.0	[1.0, 3.0]	—	**fit**
V_CE	1.7	—	V	**frozen** (§20.5)
R_contact	1×10^{-4}	—	Ω·m²	**frozen** (§20.5)
All transport (δ, κ_eff, D_mix, m_i, W, L, ν, σ, ρ)	—	—	—	**frozen**

Total fit dimension: 9 (was 6 in v6, 8 in v6.x). Core has 48 rows × 3 species = 144 residuals — 16× overdetermined.

```
# v7 theta layout in fit_kinetics.jl
const N_THETA = 9
const THETA_LB = [-6.0, -6.0, -6.0,    0.30, 0.30, 0.30,    1.0, 0.5, 1.0]
const THETA_UB = [-1.0, -1.0, -1.0,    0.70, 0.70, 0.70,    3.0, 2.0, 3.0]
const THETA0   = [-3.0, -3.0, -3.0,    0.50, 0.50, 0.50,    2.0, 1.0, 3.0]
                # j0    j0    j0       ac    ac    ac       n     n     n
                # ADN   PN    TCH      ADN   PN    TCH      ADN   PN    TCH
# HER values (j0_3 = 2.666e-5, alpha_c_3 = 0.390) are frozen at v6.x converged
↪  values
# and applied permanently via KIN_OVERRIDE before each fixed-j solve.
```

> **Open question for n_ADN bound.** The v6.x first fit returned n_ADN = 1.000 @ LB. v6.x §9.6 Finding 1 argued for relaxing this bound to 0.5 to *diagnose* whether the bound-pinning is driven by Langmuir-Hinshelwood saturation or by missing TCH species. v7 *now has TCH*, so the diagnostic experiment is: keep `THETA_LB[7] = 1.0` for the first v7 fit and check whether n_ADN comes off the bound naturally. If it still pins at 1.0 with TCH present, *then* relax to 0.5 in a v8 fit and check coverage saturation.

20.4 Targets

Metric	Target	Notes
Stage 4a — Core		
FE_ADN RMSE on Core	< 8 pp	GPR surrogate gets 2.2–2.8 pp
FE_PN RMSE on Core	< 5 pp	—
FE_TCH RMSE on Core (NEW v7)	**< 4 pp**	**Tighter than PN — TCH FE values span a smaller range (5–17%)**
Core FE_ADN at j = 100–150 mA cm^{-2}, ε_org = 0.15, gap = 0.5 mm	> 70%	Anchor rows
Stage 4b — Extended (forward apply, no re-fit)		
FE_ADN RMSE on Extended	< 12 pp	If exceeded -> bubble convection matters at high j
Residual structure	random vs (j, ε_org)	Systematic vs j => kinetic saturation; vs ε_org => D_mix
Stage 4b — Full holdout (forward apply)		
FE_ADN RMSE on Holdout	< 15 pp	If exceeded -> bubble work mandatory in v8
Holdout – Extended RMSE delta	< 5 pp	The "gap-dependent bubble penalty" magnitude

20.5 Why V_CE and R_contact are frozen in v6

Bloomquist's SI tables omit per-row V_cell directly. V_cell can be back-derived from `EP_ADN` (energy productivity, kg kWh^{-1}) but the reverse map carries compounded measurement noise from PR_ADN, j, and EP_ADN. Fitting V_CE and R_contact against a back-derived noisy quantity would *reduce* the trustworthiness of those parameters, not increase it. Two cleaner options for v7:

1. Re-acquire per-row V_cell from a future experimental refresh (Bloomquist's group has the raw data).
2. Couple V_CE and R_contact to the bubble physics (since η_anode and ε_gas both depend on j) and fit the combined object once v7 lands.

Defaults used in v6: `V_CE = 1.7 V` (= 1.23 V OER thermo + 0.45 V SS overpotential at ~100 mA cm^{-2}, lit average), `R_contact = 1×10⁻⁴ Ω·m²` (typical spring-probe + Cd-foil contact stack).

20.6 What if the kinetics-only fit fails?

If Stage 4a Core FE_ADN RMSE exceeds 12 pp, the kinetics form is too rigid — likely the second-order AN dependence on j_1 is not enough to explain the FE-vs-ε_org curve at moderate j. Diagnostic protocol:

- **Residuals correlate with j** -> α_c too small / kinetic saturation needed.
- **Residuals correlate with ε_org** -> D_mix arithmetic mean is too weak; upgrade to m_i correction (§4.2) before re-fitting.
- **Residuals correlate with FE_TCH** (compute from Bloomquist column) -> TCH is sucking up current the model attributes to ADPN. Add TCH species (§19.3) before re-fitting.

In all three cases the answer is to fix the model, *not* to relax the fit bounds. v6 forbids “softening” α_c bounds beyond [0.3, 0.7] — values outside that range have no physical Tafel interpretation.

20.7 Module: `fit_kinetics.jl`

```
module FitKinetics
using ..Solver, ..CellVoltage, ..Hydro, ..Chemistry, CSV, DataFrames

const V_CE_FROZEN     = 1.7    # V vs SHE
const R_CONTACT_FROZEN = 1e-4  # Ω·m²

function select_core(df)
    return filter(df) do r
        r.gap_mm in (0.5, 1.0) && r.j_mA_cm2 <= 190 && r.phi_AN >= 0.04
    end
end
select_extended(df) = filter(r -> r.gap_mm in (0.5, 1.0) && r.phi_AN >= 0.04, df)
select_holdout(df)  = filter(r -> r.gap_mm == 0.25 && r.phi_AN >= 0.04, df)

function loss_core(theta, df_core)
    j0_1, j0_2, j0_3, ac1, ac2, ac3 = theta
    res_sq = 0.0
    for row in eachrow(df_core)
        gap_m   = row.gap_mm * 1e-3
        Q_tot   = Hydro.ml_min_to_m3_s(row.Q_total_mL_min)
        delta   = Hydro.delta_leveque(gap_m, Q_tot)
        eps_org = row.phi_AN
        j_target = row.j_mA_cm2 * 10.0       # mA/cm² -> A/m²
        FE_ADN_model, FE_PN_model, _ = solve_at_j(
            j_target, eps_org, delta;
            j0=(j0_1,j0_2,j0_3), alpha_c=(ac1,ac2,ac3))
        res_sq += (FE_ADN_model - row.FE_ADN_pct)^2
        res_sq += (FE_PN_model  - row.FE_PN_pct)^2
    end
    return res_sq
end
end
```

---

21. Required Plots

Per-voltage profile figures use a **2×3 grid** to accommodate the pH panel:

Panel	Content	Axes
(0,0)	H^+ and OH^- (log y)	x [μm] / c [mol m^{-3}]
(0,1)	Phosphate speciation	x [μm] / c [mol m^{-3}]
(0,2)	**pH = $-\log_{10}$(c_H/1000)** (with bulk reference line)	x [μm] / pH
(1,0)	AN / ADPN / PN	x [μm] / c [mol m^{-3}]
(1,1)	ϕ_l	x [μm] / mV
(1,2)	(reserved)	

Per-sweep diagnostic panels:

Panel	Content	Axes
(a1)	j_ADPN, j_PN, j_HER vs V — **log y**	–V vs SHE / mA cm^{-2}
(a2)	j_ADPN, j_PN, j_HER vs V — **linear y**	–V vs SHE / mA cm^{-2}
(b)	FE_ADPN, FE_PN, FE_HER vs V	–V vs SHE / %
(c)	FE_ADPN vs j at multiple ε_org	j / FE — key Bloomquist comparison
(d)	FE_ADPN vs ε_org at fixed j	ε_org / FE
(e)	c_AN(0)/c_AN,bulk vs j	j / depletion ratio
(f)	ϕ_l(0) vs j at multiple ε_org	j / mV (ohmic penalty)
(g)	D_AN,mix and D_OH,mix vs ε_org — **show regime transition at ε_sat**	ε_org / D
(h)	Production rate vs ε_org	ε_org / kg cm^{-2} h^{-1}

Panel (g) is the most important diagnostic of the regime transition — it shows the step at ε_sat where single-phase D_aq gives way to the two-phase arithmetic mean.

v6 fit-validation panels

Panel	Content	Axes
(i)	**FE_ADN model vs measured** parity, color by gap	measured / model FE [%]
(j)	**FE_PN model vs measured** parity, color by gap	measured / model FE [%]

Panel	Content	Axes
(k)	FE_ADN residual vs j, faceted by gap	j [mA/cm²] / Δ FE_ADN [pp]
(l)	FE_ADN residual vs ε_org, faceted by gap	ε_org / Δ FE_ADN [pp]
(m)	Lévêque δ vs (gap, Q_total) — surface	gap / Q / δ [μm]
(n)	κ_eff vs ε_org with κ_dilute and Bruggeman cutoff lines	ε_org / κ [S/m]
(o)	We_aq vs We_org regime map with Bloomquist points overlaid	We_aq / We_org

Panels (i)–(l) are the canonical fit-quality plots; the 0.25 mm gap should appear as a clear outlier band in (k) and (l) if the kinetics fit is correct and bubble physics is the missing piece.

v7 additional fit-validation panels (NEW)

Panel	Content	Axes
(p)	**FE_TCH model vs measured** parity, color by gap	measured / model FE_TCH [%]
(q)	FE_TCH residual vs j, faceted by gap	j [mA/cm²] / Δ FE_TCH [pp]
(r)	FE_TCH residual vs ε_org, faceted by gap	ε_org / Δ FE_TCH [pp]
(s)	4-panel parity composite: FE_ADN, FE_PN, FE_TCH, V_cell	each measured / model
(t)	Loss trajectory (LM iterations): loss, λ, accept/reject	iter / loss (log)

The 4-panel composite (s) is the canonical figure for v7 fit reports — pulls all three FE channels plus the V_cell parity onto one page for direct comparison with v6 / v6.x predecessors.

---

22. v6 -> v7 Changelog Summary (NEW v7)

This section consolidates the deltas between v6 (the canonical baseline retained in `an_ehd/`) and v7 (`an_ehd_v2/`), incorporating both the v6.x patches (n_ADN, n_PN as fit parameters — see CHANGELOG_V5toV6 §9) and the v7 TCH additions in this guide. Every change here is a code-and-spec change; feature additions only.

22.1 Structural changes

	v6	v6.x	**v7**
n DOF species	8	8	**9** (TCH appended)
DOFs per cell (species + ϕ_l)	9	9	**10**
Total DOFs (N_mesh = 100)	900	900	**1000**
Banded Jacobian halfbw	17	17	**19**
Banded FD colors	35	35	**39**
Cathodic reactions	3	3	**4** (j_TCH added)
KIN_OVERRIDE shape	(j0:NTuple{3}, ac:NTuple{3})	(j0:3, ac:3, n:NTuple{2})	**(j0:NTuple{4}, ac:NTuple{4}, n:NTuple{3})**
`tafel_currents` return	3-tuple	3-tuple	**4-tuple**
AN reaction orders	hardcoded n_ADN=2, n_PN=1	fit params via override	fit params + n_TCH (initial 3)
Fit dimension N_THETA	6	8	**9**
HER kinetics	fit	fit	**frozen at v6.x converged values**
FE_TCH	not modelled	not modelled	**fit residual channel**

22.2 File-by-file delta

File	v6 -> v7 changes
`params.jl`	Extend D_aq, D_org, m_partition, z_species, c_bulk arrays to length 9 (TCH appended). Bump n_species, JAC_BLOCK, JAC_HALFBW, n_colors. Add j_0_TCH, α_c_TCH, n_TCH_default, E°_TCH, n_e_TCH constants. Update HER constants to v6.x converged values.
`kinetics.jl`	KIN_OVERRIDE Ref -> 4-tuple j0/ac, 3-tuple n. `tafel_currents` returns 4-tuple incl. j_TCH = j_0_TCH (cA/c_ref)^n_TCH exp(-α_TCH F η_TCH/RT).

File	v6 -> v7 changes
`assembly.jl`	DOF index `9·(ix−1)+k -> 10·(ix−1)+k` for species, `10·ix` for ϕ_l. Faradaic flux BCs updated for 4 reactions: J_AN(0) = $-(2j_1+j_2+3j_4)/(2F)$; J_TCH(0) = $+j_4/(2F)$; J_OH(0) = $+(j_1+j_2+j_3+j_4)/F$.
`chemistry.jl`	`make_initial_guess` and `bulk_concentration` populate index 9 (c_TCH = 0).
`diffusivity.jl`	`D_mix` arrays handle length-9 species. Verify `1:n_species` parametrisation.
`transport.jl`	Audit `1:n_species` loop bound; no logic changes expected.
`fixed_j_solver.jl`	`solve_at_j` accepts j0::NTuple{4}, alpha_c::NTuple{4}, n_orders::NTuple{3}. Default n_orders = (2.0, 1.0, 3.0).
`fit_kinetics.jl`	N_THETA = 9. New layout per §20.3. HER frozen via `J0_3_FROZEN`, `ALPHA_C3_FROZEN` constants applied permanently via KIN_OVERRIDE before each fixed-j solve. Residuals: F length is `3·length(sel)` with FE_ADN, FE_PN, FE_TCH stacked per row.
`run_stage4v3.jl`	NEW. Mirrors v6.x `run_stage4v2.jl` but writes to `an_ehd_v2/output/stage4v3/`.
`run_stage4.jl`, `run_stage4v2.jl`	Both gated with `error()` (in an_ehd/) to prevent overwriting v6 / v6.x baselines.
`analyze_stage4.jl`	Read 9-key fitted-theta and emit `stage4v3_diagnostic.csv` with FE_TCH column.
`plot_stage4v3_*.py`	Target `an_ehd_v2/output/stage4v3/`. Add FE_TCH parity panel. FE_HER residual now uses corrected `100 − FE_ADN − FE_TCH −` FE_PN denominator (no longer absorbs TCH silently).

22.3 Cache invalidation

The 9-species DOF layout (1000 unknowns, 10 DOFs/cell) invalidates every v6 `output/cache/` binary file (which is 9 DOFs/cell). v7 must re-run **Stages 1, 2, 2m, 3** in `an_ehd_v2/` before Stage 4v3.

Estimated wake compute:

Stage	Wall time	Notes
Stage 1 (single point)	~2 min	Smoke test of 9-species solver
Stage 2 (ε_org sweep)	~10 min	Confirms v6 ε_org trends survive 9-species transition
Stage 2m (m_i-corrected)	~10 min	Optional unless m_i upgrade revisited
Stage 3 (warm-start cache)	~15 min	Required before Stage 4v3
Stage 4v3 (TCH-aware fit)	~75 min	9 fit params, 48 Core rows × 3 species channels

Total: ~2 hours wake compute. Use `caffeinate -dimsu -w <julia_pid>` and stay on AC; lid-closure on battery suspends Julia even with caffeinate running (the `-s` flag is documented as AC-only on macOS).

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23. v7 -> v8 Roadmap (NEW v7)

After v7 lands and produces a Stage 4v3 fit, the structural physics most likely to remain unresolved is bubble-induced convection in the cathode boundary layer. v8 priorities, ordered by expected residual reduction:

1. **`f_bubble(j, gap, Q)` enhancement on δ_lev** (cf. v6 §7 Step 6c). Closes the 0.25 mm gap Holdout gap. Empirical correlation form `δ_actual = δ_lam · (1 + α_b · (j/j_ref)^β_b)`; one or two new fit scalars depending on whether α_b / β_b are both fit or just α_b.
2. **Bruggeman void factor on κ_eff** (`κ_eff = κ_dilute · (1−ε_org)^1.5 · (1−ε_gas)^1.5`). Helps V_cell parity at high j and small gap.
3. **`ε_gas(j, gap, Q)` model.** Faraday + residence-time as the first cut: `Q_H₂ = j·A/(2F)`, `τ_residence = L_channel / v_super`, `ε_gas = Q_H₂·τ/V_channel × correction_factor`. Promote correction_factor to fit param if needed.
4. **(conditional) n_ADN lower bound relaxed to 0.5** — only if v7 still pins n_ADN at LB after TCH lands. That would point at coverage saturation rather than missing-species, and motivates a different transport-side investigation.
5. **(conditional) Re-fit V_CE, R_contact** — only after bubbles are in. Frozen v6.x values are expected to be within 0.2 V of optimal; refitting only buys tiny accuracy improvement before bubbles land.
6. **(conditional) Per-channel σ weights on the loss** (`σ_ADN ~= 5 pp`, `σ_PN ~= 2 pp`, `σ_TCH ~= 3 pp` from Bloomquist GPR surrogate). Rebalances the joint fit if v7 trades excessive PN accuracy for ADN.

Stop condition

Don't add parameters without a residual diagnostic that justifies them. The instruction-follow heuristic from CHANGELOG_V5toV6 §7: stop adding parameters when (1) residuals look random vs (j, ε_org, gap), (2) Core combined RMSE is in the noise floor (~5 pp matching σ_FE,ADN), and (3) the next param doesn't reduce parity-plot scatter visibly.

References: Bloomquist et al. CEJ 2026; Corpus et al. Joule 2023; Weng, Bell & Weber PCCP 2018; Huang et al. CEJ 2020; Suwanvaipattana et al. J. Cleaner Prod. 2017; Mathison et al. JACS 2025; Costentin & Savéant J. Electroanal. Chem. 2004; Lasia J. Electroanal. Chem. 1995; Eigen & De Maeyer Z. Elektrochem. 1955; Newman, Electrochemical Systems 3rd ed.; Bird/Stewart/Lightfoot Transport Phenomena 2nd ed.; Lévêque, Ann. Mines 1928; Trasatti J. Electroanal. Chem. 1972 (HER on Cd literature comparison).

Guide v5 written 2026-04-21. Primary changes from v4: corrected AN bulk formula (Convention A, per total volume), regime-aware D_mix, OH^--pathway phosphate buffer chemistry with Eigen–De Maeyer water rates, ForwardDiff AD Jacobian option with Taylor-smoothed Bernoulli branch, direct `(J+λI)`du=−F Newton with tol=10^{-5}, ε_org sweep shifted to [0.02, 0.30] (never 0). See CHANGELOG_V4toV5.

Guide v6 written 2026-04-27. Primary changes from v5: external cell-voltage decomposition (§17); dilute-solution κ_eff(c_bulk, ε_org) computed from model state, no fit param (§17.2); Lévêque-only δ <-> flow mapping (§18, no bubble correction); Bloomquist 162-row dataset (§19); kinetics-only fit on 6 kinetic parameters with all transport frozen (§20); Stage 4 added (§12); fit-validation panels (§21 i–o). Bubble void corrections, K_δ geometric factor on δ, and TCH species deferred to v7. See CHANGELOG_V5toV6 for design rationale.

Guide v7 written 2026-04-28. Primary changes from v6: TCH (1,3,6-tricyanohexane) added as species 9 with its own Tafel reaction (n_e = 2 per Bloomquist SI; §2, §5); AN reaction orders n_ADN, n_PN, n_TCH promoted from hardcoded constants to fit parameters via the extended `KIN_OVERRIDE` Ref (§5.2, §14.1); HER kinetics frozen at v6.x converged values (§9.2, §20.3); fit dimension extended to N_THETA = 9 (§20.3); FE_TCH added as a residual channel; Faradaic flux BCs updated (§7.1); v7 model code lives in `an_ehd_v2/` (§14); bandwidth 17 -> 19 / colors 35 -> 39 (§10.2). Bubble physics still deferred to v8 (§23 roadmap).